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Assessing the levelized cost of energy in South Korea

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ABSTRACT

This study evaluates the levelized cost of energy (LCOE) for various energy technologies in the Republic of Korea (Korea) from 2023 to 2050, highlighting cost trajectories and potential crossovers among competing technologies. The analysis projects that, based on our set of assumptions, utility-scale photovoltaic systems achieve lower LCOEs than nuclear by 2030, while fixed offshore wind is expected to become cost-competitive with coal-fired generation around the same time. Floating offshore wind is projected to reach cost parity with coal in the late 2030s. Co-firing with natural gas and green hydrogen is identified as the highest-cost generation option due to high natural gas and green fuel costs and declining capacity utilization. This study further examines the potential for hybrid systems that integrate renewable energy with energy storage to serve as flexible, cost-effective, zero-emission alternatives to green hydrogen-based generation. Spatial LCOE assessments indicate that near-shore offshore wind sites may achieve lower costs despite modest capacity factors, contingent on site-specific factors such as grid integration and social acceptance. The findings indicate that renewable energy technologies are expected to experience continued cost declines, with solar photovoltaic becoming the most competitive energy source in Korea by 2030–2035. Incorporating social costs accelerates this shift from conventional alternatives.

1. Introduction

In the Republic of Korea (i.e., South Korea, hereafter referred to as "Korea"), long-term shifts in the energy sector are increasingly shaped by domestic resource constraints, energy security considerations, and the evolving economics of power generation technologies. Korea's heavy dependence on imported fossil fuels and its currency volatility against the U.S. dollar both call for urgent action to ensure an affordable and resilient energy future for Korea and its citizens. Long-term fuel price trends in Korea are highly uncertain, as oil and gas prices spiked to record levels in 2011 and again in 2022—in the latter case, rising by 109 % compared to average prices over the previous decade. In 2020, with only 7 % of Korea's primary energy coming from domestic resources, liquefied natural gas (LNG) and coal power plants provided nearly two-thirds of Korea's total electricity generation [1]. At the time of this study, the Korean won-U.S. dollar exchange rate had soared to around 1480 won per dollar, a level not seen since the global financial crisis of 15 years ago [2,3].

In 2021, Korea enacted the Carbon Neutral Framework Act, which called for carbon emissions reductions of at least 35 percent below 2018 levels by 2030; under the law, the government then set a 40 percent reduction goal. Korea's 11th Basic Plan for Electricity Supply and Demand[1] ("The 11th Basic Plan") anticipates that by 2038, solar and wind will reach approximately 118 GW (GW) of generating capacity and contribute about 27 % to the country's electricity generation mix [4]. Achieving these targets will require the accelerated deployment of cost-competitive energy technologies, as global market dynamics shift and technologies continue to evolve over time [5–7]. Globally, deployment of renewable energy (RE) technologies has accelerated rapidly since 2019, outpacing conventional fossil-based technologies by a two-to-one ratio [8]. From 2022 to 2023, global installed capacity of solar photovoltaic (PV) and wind increased by 85 % and 60 %, respectively [8]. Furthermore, global installed RE capacity is projected to nearly triple by 2030, with strong market momentum shown in many regions. This trend reflects broad advances in RE technology performance, manufacturing scale, and cost reductions.

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[1] Korean government addresses comprehensive documentation of its electricity regulations and plans, including foundational principles of electricity supply and demand, long-term forecasts, power system planning, and demand-side management. The Basic Plan is updated every two years, most recently in March 2025 (the 11th Basic Plan) covering the period 2024 to 2038.

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Reliable cost projections of future energy technologies play a crucial role in informing long-term electricity sector planning and investment decisions. To serve this need, this study applies a levelized cost of energy (LCOE) framework to evaluate the projected costs of key energy generation technologies in Korea through 2050. While LCOE is a widely used metric, its application to Korea's energy transition—incorporating domestic cost trends, fuel price uncertainties, technology maturity, and policy-relevant assumptions—offers timely and actionable insights that are deeply relevant to the country's energy future.

This study extends the conventional LCOE approach by incorporating social costs and by evaluating the prospective role of co-combustion with eco-friendly fuels such as green ammonia and green hydrogen. The analysis contributes to the literature in three key ways. First, it provides a comparative assessment of LCOE and "social LCOE" (i.e., incorporating social costs) across multiple technologies and scenarios through 2050, identifying when cost shifts occur and which resources gain advantage over time. Second, it explores the competitiveness of co-firing with green fuels relative to solar-plus-storage hybrid systems as potential non-emitting balancing options. Third, it presents a spatially detailed offshore wind LCOE analysis, highlighting regional generation potential and interconnection opportunities. Collectively, these components offer policy-relevant insights for Korea's energy future while providing a framework that can inform similar efforts globally.

This paper is organized as follows. Section 2 introduces the LCOE framework and its relevance to technology and policy evaluation. Section 3 outlines the study scope, data sources and methodology, including key assumptions. Section 4 presents LCOE projections across technologies and scenarios, along with sensitivity analyses and a discussion of key drivers. Section 5 concludes with insights for Korea's power system planning.

2. LCOE overview

Prioritizing cost-competitive generation investments in the power sector requires evaluating future technology options. To facilitate energy technology investment decisions, simplified metrics such as LCOE are commonly employed. LCOE is a standardized metric used to evaluate and compare the economic viability of diverse energy generation technologies. It is calculated by determining the present value of all capital and operational expenditures associated with an energy source throughout its operational lifespan, subsequently dividing this total by the projected energy output. This methodology facilitates direct comparisons between varying technologies such as wind, solar PV, and natural gas, irrespective of differences in operational lifespan, project scale, initial investment, risk profile, expected return, or generating capacity.

By projecting life-cycle costs, LCOE provides a long-term perspective on economic performance, enabling data-driven decisions regarding project value and feasibility. For decision-makers in municipalities or regional planning agencies, LCOE analysis can identify opportunities for developing projects at various scales—ranging from facility-level installations to community-wide or commercial ventures—and provides a basis for economic comparison against existing utility rates. The LCOE methodology has been applied to a wide variety of energy sources, including solar PV [9,10], electrical energy storage [10], nuclear [11], wind [12,13], wave and tidal [14], biomass [15,16], and hybrid systems combining PV, batteries, and cogeneration [17]. Developers can use LCOE to determine generation costs during electricity sales negotiations to ensure they receive a price that exceeds costs.

LCOE is a summary metric that integrates key technology cost and performance parameters to facilitate the comparison of diverse energy resources. These parameters include.

- Overnight capital costs (OCC): The upfront expenses required to build the resource and associated infrastructure, including grid connection costs.
- Fixed operations and maintenance (FOM) costs: Ongoing costs of maintaining and operating the resource, regardless of output.
- Variable operations and maintenance (VOM) costs: Costs that fluctuate based on the resource's output.
- Fuel costs: The cost of fuel needed to operate the resource. (Not applicable where "fuel" is the sun, wind, or water.)
- Financing costs: The expense of borrowing money to build the resource.
- Capacity factor (CF): The percentage of a resource's actual output relative to its maximum possible output, assuming continuous operation at nameplate capacity over the same period.

LCOE is particularly valuable for evaluating the potential impacts of technological advancements under various technology scenarios, as it synthesizes the combined effect of key cost and performance parameters. While LCOE is widely used for comparing energy generation technologies and setting research and development priorities, it has well-documented limitations when used to assess the competitiveness of energy technologies, primarily because it does not account for a technology's broader economic value within the energy system. For instance, LCOE can overlook critical factors such as temporal and locational grid impacts, ancillary service benefits, or the ability to complement intermittent renewable resources. These factors can arise from cost assumptions about environmental impacts, dispatchability, interest rates, and other issues [18–22].

2.1. Alternative metrics

To address the well-known limitations of LCOE—such as its inability to capture temporal value, locational variation, system integration costs, and social externalities—numerous improvements and complementary approaches have been proposed. For instance, some studies suggest incorporating correction factors to reflect the intermittency costs of renewables [23], while others highlight the static nature of LCOE and advocate for probabilistic modeling approaches to better capture uncertainty [24–26]. These critiques are echoed in recent literature, including a comprehensive review by Emblemsvåg [27] which cautions that LCOE, if used in isolation, can mislead policy and investment decisions.

In response, several alternative or extended metrics have been developed, such as the Levelized Avoided Cost of Electricity (LACE), Levelized Value of Electricity (LVOE), Net Value of Electricity (NVOE), and System LCOE [22,28,29], all of which seek to reflect the broader system value and integration dynamics of power generation technologies. The International Energy Agency's Value-Adjusted Levelized Cost of Electricity (VALCOE) metric, for example, incorporates energy, capacity, and flexibility contributions into a single indicator [30], while the Levelized Cost of Storage (LCOS) has emerged as an analogous metric for energy storage technologies, capturing both charging and discharging economics [31]. Although these expanded metrics offer a more comprehensive perspective, we adopt LCOE in this study for its comparability and transparency, and complement it with spatially differentiated estimates and explicit incorporation of social costs tailored to the Korean context. Further discussion on alternative metrics is provided in Appendix D.

2.2. LCOE literature review

LCOE has been extensively used in energy economics and engineering to compare generation costs across technologies and countries. Numerous studies have examined its assumptions and limitations, including sensitivity to discount rates and capacity factors [32]. Technology-specific LCOE studies further illustrate the diversity in costs.

For instance, PV costs vary by cell type and deployment context [9,33], and Thai and Brouwer [34] examined the cost implications of transmitting electricity from remote utility-scale PV sites, comparing electric transmission and hydrogen pipeline delivery. Arenas-López and Badaoui [35] analyzed the LCOE of offshore wind by accounting for wind speed uncertainty and provided estimates of the resulting uncertainty range.

In addition, spatial LCOE mapping has become a critical tool for identifying least-cost deployment zones by accounting for geographic variation in resource availability, expected energy yield, and infrastructure—for tidal stream energy [36], offshore winds [37,38], and land-based wind and PV [39]. These studies highlight that renewable LCOEs can vary widely even within a single country, primarily due to differences in expected generation output. In the case of offshore wind, the additional cost of subsea transmission required to connect to onshore grids further amplifies this spatial variation.

LCOE is frequently used to compare generation costs across different energy resources and jurisdictions. A notable study by Ram et al. [40] conducted a comparative analysis of electricity generation costs—including renewables, fossil fuels, and nuclear power—for G20 countries over the 2015–2030 period. Within the EU, studies have used LCOE, among other metrics, to assess the cost-effectiveness of renewable energy and to inform national and regional targets [41]. The International Energy Agency's (IEA) [42] World Energy Outlook 2024 similarly emphasizes that regional differences in resource quality, infrastructure, and policy environments are key factors shaping the cost-competitiveness of clean energy technologies. In general, such international comparisons apply consistent methodologies to explain why costs differ across countries—whether due to differences in natural resource availability, labor and capital costs, or institutional frameworks. These insights provide an important backdrop for understanding how a given country fits into the global energy transition landscape.

Several studies have incorporated externalities—i.e., societal costs or benefits from power generation that are not reflected in market prices—into cost assessments of power and energy systems. Kabeyi and Olanrewaju [43] identified the exclusion of externalities as a key limitation of the traditional LCOE framework. One prominent line of inquiry has focused on integrating environmental and health-related damages into LCOE. Rhodes et al. [44] estimated LCOE across U.S. counties by accounting for greenhouse gas emissions and local air pollutants (SO2, NOx, PM2.5, and PM10), including those emitted during construction and decommissioning. Similarly, Zvingilaite [45] incorporated human health externalities in the optimal operation of Denmark's power and heat integrated energy system. Building on this, Moslehi and Reddy [46] developed a location- and time-specific Life Cycle Assessment (LCA) methodology that quantifies environmental externalities based on regional grid mix and diurnal variations, enabling real-time evaluation of the sustainability of integrated energy systems.

In the context of South Korea, Cho et al. [47] compared the LCOE of coal, natural gas, and nuclear power while incorporating social costs, including CO_2 emissions, accident risk, and air pollution damage. The International Energy Agency [42], in its World Energy Outlook 2024, presented LCOE estimates under various carbon pricing scenarios across multiple countries. Samadi [48] broadened the scope of externalities by considering factors such as carbon emissions and noise, landscape, and ecosystem impacts. Although nuclear accident risks were discussed, they were not quantified due to wide-range of uncertainties.

These studies collectively underscore the value of embedding externalities into LCOE frameworks to better capture the societal costs and benefits of electricity generation. Following this approach, this study incorporates carbon emissions, accident-related risks, and air pollution damages into the social LCOE calculation, allowing for the identification of cost-effective technologies when externalities are quantified and internalized.

2.3. Recent LCOE trends in global and Korean contexts

According to an IEA study based on 2020 data from 24 countries, renewables are highly cost-competitive overall, but in some countries (e.g., Japan, Korea, and Russia) their LCOE still exceeds that of fossil- or nuclear-based generation [49]. However, the aggregated data for these 24 countries has limitation in fully capture the complexity of levelized generation costs which can differ significantly in each country due to varying site conditions for RE generation, fuel costs, capacity factor, and technology maturity.

Globally, RE costs have fallen significantly and are expected to continue doing so, while costs for traditional energy sources are expected to remain stable or increase as CFs decrease [50].

Bolinger et al. report that in the U.S., the LCOE for wind and solar PV has converged, with both technologies exhibiting a period of learning and associated cost declines of 15 % (wind) and 24 % (solar) through 2020 [51]. With a higher learning rate and greater deployments, LCOE for solar PV and wind could drop further still.

In Korea, a recent study projects LCOE for RE technologies will decrease while coal, gas, and nuclear costs increase [52,53]. Solar PV and land-based wind are expected to be the most cost-competitive resources by 2050, while coal and gas will be least cost-competitive due to emissions and environmental costs. Notably, these previous studies did not fully account for time-varying factors such as fuel prices, CFs of conventional generators, and co-firing ratios. Furthermore, while other studies have analyzed the LCOE of offshore WT, their scope has been limited.

This study builds on prior research and global and domestic trends by providing the following key contributions.

- A comparative analysis of the LCOE and social LCOE of power generation resources being considered in Korea, under multiple scenarios and through 2050. Based on this, the study identifies when a "takeover" between resources occurs and determines which technologies become more competitive over time from an LCOE perspective. While the analysis incorporates Korea-specific inputs—such as trends in construction costs, construction periods by generator type, and decreasing CFs of fossil-fueled generators—the observed dynamics can be applied to other countries undergoing similar power sector transitions.
- LCOE comparison of green hydrogen (GH) and renewable-battery hybrid systems. In Korea, GH is considered a low-emission flexibility resource for the future. This study explores various GH co-firing scenarios and compares their LCOE with that of hybrid systems, revealing the competitive potential of the latter.
- A spatially detailed comparison of offshore wind LCOE by technology type (fixed or floating) and region. This includes estimating the potential generation and estimated power demand based on LCOE distributions at each point of interconnection.

3. Scope, data, and method

In this analysis, we consider technological advancements and future projections for energy technologies in Korea as they impact cost and performance parameters. We standardize key parameters and assumptions between the U.S. and Korea, ensuring consistency in methodology. Our methodology and key assumptions are based on the Annual Technology Baseline (ATB) scenarios developed by the National Renewable Energy Laboratory (NREL) [54]. The ATB framework emphasizes defining primary cost and performance parameters for use in electric sector modeling and other analyses requiring sophisticated comparisons.

3.1. Scope of technology coverage

This study evaluates a wide range of generation technologies

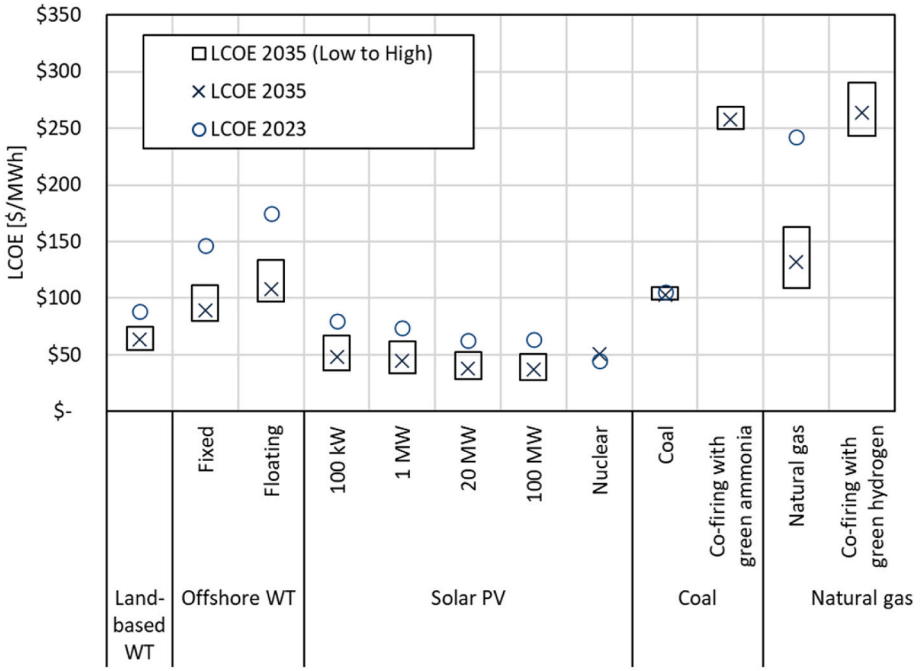


Fig 1. LCOE by generation resource in 2023 and 2035

including resources under discussion as future options in Korea. The classification below provides an overview of the technologies analyzed in this study.

- Conventional generators. This category includes coal-fired, natural gas-fired, and nuclear power plants.
- Solar PV. Solar PV systems are classified into four capacity-based categories: 100 kW, 1 MW, 20 MW, and 100 MW.
- WT. The LCOE of WT is analyzed across three technology types by installation location: land-based, fixed offshore, and floating offshore WT. In Korea, floating offshore WT remains an emerging technology with limited empirical assessment.
- Green ammonia (GA) and GH co-firing. Co-firing GA with coal and GH with natural gas are considered potential future balancing resources that support decarbonization and utilize existing infrastructure. In this study, GA and GH refer to ammonia and hydrogen produced using energy from renewable sources (i.e., solar or wind). As a baseline, we assume imported GH, but we also review a wide range of GH costs to capture the uncertainty, covering both imported and domestically produced GH.
- Renewable-battery hybrid power plants. As the costs of RE and batteries decline, hybrid plants integrating renewables and energy storage systems (ESS) may become promising flexibility resources. We compared the LCOE of hybrid power plants with GH-fired generators in terms of their role as flexibility resources.

3.2. Overview of ATB's LCOE calculation

The 2023 Electricity ATB used in this study presents cost and performance metrics for typical electricity generation plants in the U.S. [54]. The methodology for calculating LCOE is detailed in Appendix A. We projected future cost parameters—such as OCC and FOM costs for RE technologies and FOM for conventional generators—primarily based on data from the 2023 Electricity ATB. For the baseline year, cost inputs specific to Korea were used.

In the ATB, to better estimate the uncertainty of future projections, three scenarios are defined: Conservative, Moderate, and Advanced. This study adopts these scenarios to reflect a range of possible cost trajectories in the future. The three terms relate to the speed of technology innovation and amount of global installations, with the Conservative scenario assuming slower innovation and fewer installations and the Advanced scenario assuming the reverse.

- Conservative scenario: Assumes historical investment trends continue, with ongoing industrial learning but minimal changes from technological innovation. Public and private R&D investment declines, resulting in technology that closely resembles today's market.
- Moderate scenario: Assumes innovations currently observed in the market become more prevalent and near-market-ready technologies gain adoption, but public and private R&D investment remains steady. This represents the expected level of technology innovation.
- Advanced scenario: Assumes technologies that are not yet market-ready achieve success and become widespread. New technology architectures may differ significantly from those seen today, with increased public and private research and development (R&D) investment driving these advancements.

Each scenario calculates the LCOE with different parameter values, using varied assumptions for costs and other inputs. In the Conservative scenario, these parameter values generate a higher LCOE compared to the Moderate scenario for the same year, while the Advanced scenario generates the lowest LCOE values.

3.3. Key parameters

In calculating LCOE, the most influential cost and performance parameters vary depending on the energy source being analyzed. Ideally, the same study scope would apply to all generators; however, variations in criteria are necessary due to insufficient sample sizes across generator types.

LCOE for generator g in year y in $/MWh, $LCOE_{g,y}$, is calculated as Eqs. (1) and (2):

$$LCOE_{g,y} = \frac{FCR_g \times CPX_{g,y} + FOM_{g,y}}{CF_{g,y} \times 8{,}760\ hours} \times 1000\text{kW} \Big/ \text{MW} + FC_{g,y} + PC_{g,y} \quad (1)$$

$$CPX_{g,y} = CFF_g \times OCC_{g,y} \quad (2)$$

where FCR_g is a fixed charge rate for generator g, the amount of revenue per dollar of investment that must be collected annually from customers to pay the carrying charges on that investment. $CPX_{g,y}$ is capital expenditure (CAPEX) for generator g in year y in $/kW, the expenditures required to bring the plant to commercial operation. $CPX_{g,y}$ is a multiplication of CFF_g, construction finance factor, and $OCC_{g,y}$, overnight capital costs which include on-site electrical equipment. $FOM_{g,y}$, fixed operation and maintenance expenses, is annual expenditures to operate and maintain equipment that are not incurred on a per-unit-energy basis, in $/kW-yr. $CF_{g,y}$ refers to net CF, the percentage ratio of actual annual output to output at rated capacity for an entire year. $FC_{g,y}$ refers to fuel costs of a generating plant in $/MWh. $PC_{g,y}$ refers to policy costs, levied by the government on generator types such as nuclear, coal, and natural gas to address environmental, economic, safety, or other concerns.

Fuel cost is an important factor when considering the possible implementation of GH and GA co-firing. To account for the co-firing, the fuel cost for generator g in year y is calculated as Eq. (3):

$$FC_{g,y} = \left(\left(1 - \theta_{g,y}^{cofire}\right) \times FC_{g,y}^{MMBTU,Conv} + \theta_{g,y}^{cofire} \times FC_{g,y}^{MMBTU,Green}\right) \times HR_g \quad (3)$$

where $\theta_{g,y}^{cofire}$ is the co-firing ratio of generator g in year y, expressed in terms of heat. $FC_{g,y}^{MMBTU,Conv}$ is the fuel cost of conventional fuels such as coal, natural gas, and nuclear, in $/MMBTU. $FC_{g,y}^{MMBTU,Green}$ is the fuel cost of green fuels such as GH and GA, which are co-fired with the conventional fuels. HR_g is the heat rate of generator g, an efficiency metric expressed in MMBTU/MWh, which varies by generator type. For nuclear power, which does not involve co-firing, $\theta_{g,y}^{cofire}$ is zero for all years.

In operating a power generation company, additional costs arise beyond OCC and fuel costs. The policy costs, $PC_{g,y}^{total}$, defined in the Korean context, include carbon costs, subsidies, and local taxes, as shown in Eqs. (4)–(6):

$$PC_{g,y}^{total} = PC_{g,y}^{Carbon} + PC_{g,y}^{LocalTax} + PC_{g,y}^{subsidy} \quad (4)$$

$$PC_{g,y}^{Carbon} = e_{g,y}^{CO_2} \times \alpha_y^{\mathrm{CERR}} \times CP_y \quad (5)$$

$$e_{g,y}^{CO_2} = e_{g,y}^{CO_2,Conv} \times \left(1 - \theta_{g,y}^{cofire}\right) \quad (6)$$

where $PC_{g,y}^{Carbon}$ is a carbon cost and $PC_{g,y}^{LocalTax}$ is a local tax on conventional generators. $PC_{g,y}^{subsidy}$ is composed of the subsidies given to the area adjacent to power plants and transmission lines (and radioactive waste facilities, in the case of nuclear power).

In Korea, coal and natural gas power plants currently pay carbon costs for only 10 % of their emissions [55]. This proportion, referred to in this study as the carbon emission responsibility ratio (CERR), is denoted as α_y^{CERR}. Carbon cost for generator g, $PC_{g,y}^{carbon}$, is a multiplication of its CO_2 emissions in kg per MWh, $e_{g,y}^{CO_2}$; the CERR; and the carbon cost per unit of CO_2 emission, CP_y. CO_2 emissions in kg per MWh is a multiplication of the CO_2 emissions factor for conventional generators, $e_{g,y}^{CO_2,Conv}$, and the portion that does not use green fuels such as GH or GA.

In this study, we adopt the concept of social LCOE, inspired by Ref. [47,52,56]. Social LCOE is an extension of the conventional LCOE framework that incorporates the social costs associated with electricity generation. These social costs include environmental impacts (e.g., carbon emissions), public health effects (e.g., air pollution), and potential accident risks (e.g., for nuclear energy)—factors that are not directly reflected in market prices but are incurred by society at regional or global levels. By accounting for these externalities, social LCOE provides a more comprehensive assessment of the economic impact of various energy technologies, offering valuable insights for policy decision making. Social LCOE is calculated by adding social costs ($SC_{g,y}^{total}$) as shown in Eqs. (7)–(9):

$$LCOE_{g,y}^{social} = LCOE_{g,y} + SC_{g,y}^{total} \quad (7)$$

$$SC_{g,y}^{total} = SC_{g,y}^{carbon} + SC_{g,y}^{pollution} + C_{g,y}^{accident} + C_{g,y}^{subsidies} \quad (8)$$

$$SC_{g,y}^{carbon} = e_{g,y}^{CO_2} \times \left(1 - \alpha_y^{\mathrm{CERR}}\right) \times CP_y \quad (9)$$

$SC_{g,y}^{carbon}$ represents the potential carbon cost for coal and natural gas power plants. As these plants in Korea currently pay for only 10 % of the carbon dioxide they emit, the remaining 90 % is included in the social cost. Air pollution cost from coal and natural gas, $C_{g,y}^{pollution}$, encompasses the monetized impact of environmental damages from air pollutants such as nitrogen oxides, sulfur oxides, and particulate matter. For nuclear plants, accident risk cost, $C_{g,y}^{accident}$, and potential subsidies for areas near radioactive waste facilities, $C_{g,y}^{subsidies}$, are incorporated.

3.4. Values for parameters

3.4.1. FCR

Fixed charge rate is the amount of required revenue per dollar of investment that must be collected annually from customers to pay the carrying charges on that investment. It is a multiplication of the real capital recovery factor (4.5–4.7 %) and project finance factor (1.076–1.081).

This analysis assumes a debt fraction (DF) of 80 % for all resources. The nominal interest rate, which varies by technology, is set at 3.8 % for conventional generators based on real-world data from Korea [57] and 4.4 % for RE following a domestic LCOE study [53]. Other financial parameters are as follows.

- Inflation rate. Over the past 10 years, Korea's consumer price index has increased by an average of 1.8 % per year [58], while the core inflation rate—which excludes items with high year-to-year variability (e.g., vegetables, fruits, gasoline, and natural gas)—has been 2.0 % [59]. This study employs the core inflation rate as the base inflation rate throughout the study period.
- Rate of return on equity (RORE). RORE is defined as the assumed rate of return on the portion of assets financed through equity. A long-term average RORE of 8.0 % is applied to all technologies and periods, consistent with the average reported for publicly listed companies in Korea over the past decade [60]. For context, the Bank of Korea's 2023 Business Management Analysis reported ROREs for electric power companies ranging from −17.6 % to 0.5 % for large enterprises, 4.4 %–6.4 % for mid-sized enterprises, and 2.8 %–13.6 % for small and medium enterprises from 2021 to 2023 [61].
- Tax rate. The tax rate is fixed at 19 % across all scenarios, reflecting the five-year average effective tax rate for utilities in Korea, including the electricity, gas, and water sectors Tax Statistics Information Service [62]. The tax rate plays a pivotal role by influencing key financial metrics, including the weighted average cost of capital (WACC), project finance factor (PFF), and accumulated interest (AI), ultimately impacting the LCOE. However, the tax rate is not applied to calculate profit-related taxes as a cost component, since taxes depend on profits and various other factors that are difficult to estimate accurately.
- WACC. Based on the DF, RORE, and interest rate, the nominal WACC is estimated at 4.0 % for conventional generators and 4.5 % for renewables. Notably, Korea applies a social discount rate of 4.5 % when assessing the present value of government projects [63].
- PFF. The project finance factor captures the effect of depreciation schedules and tax treatments across technologies. It ranges from 1.076 for traditional generators to 1.081 for renewables. A straight-line depreciation method is applied to electric generators in Korea,

depreciated at 5 % annually over 20 years [64]. Both the tax rate and depreciation method are assumed constant across scenarios.

3.4.2. Construction finance factor

The construction financing factor refers to the share of the total capital cost allocated to financing during the construction phase, which increases as construction duration lengthens. This value varies by energy resource: from 1.027 for PV systems with a 1-year construction period to a high of 1.392 for nuclear, with the latter reflecting the approximately 10-year construction period of the recently built Saeul Nuclear Power Plant [65].

3.4.3. OCC in baseline year and future projections

OCC for the baseline year (2023) is derived in this study using Korea-specific data from multiple references [52,53,56,57].

- OCCs of $1541/kW and $1154/kW for coal and natural gas, respectively, are averaged based on data points from two references. For generators built in 2017 or earlier, construction costs from the Korean LCOE study [56] are inflation-adjusted to convert to 2023 prices. For generators built in 2018 or later, we used data provided by an office of the National Assembly of Korea [57].
- OCC of $3050/kW for nuclear in 2023 is based on costs derived from data on recently constructed Advanced Power Reactor 1400 MW units in Korea [52].
- OCCs of $1162/kW to $1190/kW for PV under 100 MW and $4985/kW for fixed offshore WT are based on the LCOE study of RE in Korea [53]. OCCs of $1001/kW for PV over 100 MW and $2053/kW for land-based WT are based on data provided by an office of the National Assembly of Korea [57]. Since no data was available for floating offshore wind in Korea, OCC for floating offshore WT is estimated to be 47 % higher than for fixed offshore WT, based on ATB data [54]. Note that in the NREL ATB, grid connection cost is listed as a separate cost item and distinct from OCC. However, due to data limitations in the referenced study [53], this study includes grid connection cost within OCC.
- For the OCC of GH co-firing with natural gas, we assumed an 8.5 % cost increase for hydrogen gas turbines compared to conventional gas turbines [66]. For the OCC of GA co-firing with coal, we assumed a $200/kW cost increase for the ammonia co-firing facility at a coal power plant [52].

Additional details on OCC in the baseline year are provided in Appendix B. OCC for future projections is based on the following assumptions.

- For the OCCs of conventional generators, annual increase rates of 1.62 % for coal and natural gas, and 2.55 % for nuclear, are assumed based on historical trends in Korea [52].
- The future average annual OCC reduction rates of 2.8 %/year and 1.8 %/year for PV and land-based WT, respectively, by 2050 are based on assumptions in the ATB [54].
- The future average annual OCC reduction rates of 2.3 %/year and 2.2 %/year for fixed and floating offshore WT, respectively, by 2050 were derived using a different approach. Since offshore WT technology is still in its infancy in Korea, simply applying ATB trends to the current-year OCC results in only limited cost reductions, and the overall cost remains high. Therefore, the OCC of offshore wind is assumed to converge with ATB projections by 2040, after which they follow the rate of change in the ATB scenarios [1].

Appendix C provides further details on the future projections of OCC for RE technologies.

3.4.4. FOM costs

- FOM costs for coal and natural gas generators in 2023, which include FOM expenses for power plants and transmission lines, are $66/kW-year and $42/kW-year, respectively [52].
- FOM costs for nuclear in 2023—which includes FOM costs for power plants and transmission lines, decommissioning funds, costs of processing low- and intermediate-level radioactive waste, and high-level waste management—total $139/kW-year [52,67].
- For RE sources, FOM costs vary: $24–34/kW-year for solar PV (depending on size); $46/kW-year for land-based WT; and $105/kW-year for offshore WT [53]. Future FOM costs for all generator types align with ATB projections [54].

Additional details on FOM costs in the baseline year are provided in Appendix B.

3.4.5. Fuel costs

Unit fuel costs in 2023 for nuclear, coal, and natural gas are $0.5, $7.6, and $21.2 per MMBtu, respectively [68]. To project the future cost of nuclear fuel, this study uses the U.S. Energy Information Administration (EIA)'s Reference Case projection, which anticipates prices will remain stable until 2050 [69]. For coal and natural gas, this study uses projections from the IEA's World Energy Outlook (WEO) [42]. Because the WEO includes prices for Japan but not Korea, we adjust Japan prices to Korean levels as detailed in Appendix C. The cost of GH and GA is assumed to be $5.7/$H_2$kg and $1.1/$NH_3$kg, respectively, in 2035, and $5/$H_2$kg and $0.9/$NH_3$kg in 2050, assuming imported GH and GA [52,70]. To consider a wide range of future GH costs, LCOE results ranging from $1/kg to $6/kg are compared in the sensitivity analysis.

3.4.6. Capacity factor

CFs for coal, natural gas, and nuclear in 2023 are assumed to be 54 %, 42 %, and 83 %, respectively, based on nationwide actual operational data from 2023 (MOTIE, 2025). In accordance with decarbonization plans for the power system, coal and gas generation is expected to decrease over time, with Korea's 11th Basic Plan projecting that coal and LNG CFs will be 42 % and 18 %, respectively, in 2038. Our analysis extends this projection through 2050. Furthermore, future plans involve co-firing GH or GA with fossil fuels, with non-co-firing generators excluded after 2040. This analytical approach, which involves using the planned CFs, is also employed by the IEA WEO [42].

CF for RE can vary by region, year, technology type, and weather conditions. To select representative values for RE technologies while accounting for this variability, we use historical data based on region-wide average values. For PV and land-based WT, CFs of 14.8 % and 20 % are used, based on data from 2018 to 2023 from resource-rich areas such as Jeolla-do and Gyeongsang-do [71]. For fixed offshore WT, we adopt a CF of 29.4 % from the Korea-specific LCOE study [52]. For floating offshore WT, since there are currently no such generators in operation in Korea, we use a potential CF of 33.0 % [1]. For future CFs, an annual increase of 0.3–0.6 % is assumed, driven by technology advancements such as improved module efficiency for PV systems and advanced controls and reduced system losses for WT [54]. Specifically, the CF for PV is calibrated to account for degradation over the 30-year cost recovery period and assumes a degradation rate of 0.5 % per year [53].

As mentioned, there can be a wide range of CF, which can be confirmed from multiple sources as shown in Table 1. Thus, we defined three scenarios: Baseline, Preferred, and Ideal. We focus on the impact of future cost reductions under the Baseline scenario and explore the further potential of RE with higher CFs under the Preferred and Ideal scenarios.

3.4.7. Heat rate

This study uses heat rates based on national average values of 9.3

Table 1
Capacity factors (CFs) by reference and scenario.

Data source	PV	Land-based WT	Fixed offshore WT	Floating offshore WT
EPSIS (2024) [71]	14.8 %	20.0 %	–	–
Lee and Lim [53,53]	15.4 %	23.0 %	29.4 %	–
Low scenario in 2035 Korea study (2025) [1]	17.0 %	19.0 %	31.0 %	33.0 %
High scenario in 2035 Korea study (2025) [1]	19.0 %	30.0 %	38.0 %	40.0 %
CF scenario	PV	Land-based WT	Fixed offshore WT	Floating offshore WT
Baseline	14.8 %	20.0 %	29.4 %	33.0 %
Preferred	17.0 %	24.5 %	34.5 %	36.5 %
Ideal	19.0 %	30.0 %	38.0 %	40.0 %

MMBtu/MWh [72], 10.0 MMBtu/MWh [72], and 9.4 MMBtu/MWh [52] for coal, natural gas, and nuclear, respectively. To account for technological advancements, the thermal efficiencies of coal and natural gas power plants are assumed to improve at a rate of 0.3 % per year between 2025 and 2035.

3.4.8. Policy cost

Carbon cost per unit of emissions follows Korea's projected trajectory as outlined in the WEO, set at \$160/CO_2-tonne in 2035 and \$200/CO_2-tonne in 2050 [42]. Again, under current regulations, Korea's coal and natural gas power plants must pay carbon costs on only 10 % of their emissions [55]. However, Korea's 4th Basic Plan for the Emissions Trading System states that the CERR for generation companies will increase substantially between 2026 and 2030 [73]. Because the plan does not specify an exact CERR target, a sensitivity analysis is conducted to examine outcomes under varying CERR assumptions for fossil-fueled generators.

A local resource usage tax of \$1/MWh is applied to nuclear, coal, and natural gas generation facilities [74]. In addition, regulatory costs and incentive payments—including subsidies for communities near power plants [75] and transmission infrastructure [76]—are incorporated at \$0.2–0.5/MWh. For nuclear power, an additional subsidy of \$0.6/MWh is applied for compensation to areas near radioactive waste facilities [77].

3.4.9. Social cost

Unpaid carbon costs (i.e., 90 % of emissions from coal and natural gas plants, under current policy) are included as a component of social cost. Air pollution costs are calculated based on monetized environmental damages from pollutants including nitrogen oxides, sulfur oxides, and particulate matter. These costs are estimated at \$10/MWh for coal, which includes all pollutants, and \$3/MWh for natural gas, which only includes nitrogen oxides [52].

For nuclear power, an accident risk cost of \$7.50/MWh is included, derived from decommissioning and compensation costs related to the Fukushima nuclear disaster and adjusted to reflect Korea's demographic and economic context [52]. This inclusion reflects the increased recognition, prompted by the 2011 Fukushima accident, of the potential cost associated with severe nuclear accidents—encompassing not only direct damages but also long-term decontamination, compensation, and administrative expenses [56]. Although such costs are typically excluded from conventional LCOE comparisons, the Fukushima disaster stands as a notable exception. Several studies focusing on Korea have included these externalities, reflecting heightened public concern and geographic proximity of population centers [47,52,56]. The \$7.50/MWh estimate is based on the global nuclear accident frequency, 2.5×10^{-4} per reactor-year, observed over cumulative reactor-year data and adjusted for Korea-specific factors such as income levels and population density based on the 2011 Fukushima case [52]. Further discussion of accident risk management cost is presented in Appendix C.

Although Korea has operated nuclear power plants for over five decades, it has not yet established a high-level radioactive waste disposal facility. Currently, low- and intermediate-level waste is stored in temporary facilities, which are projected to reach capacity around 2030 [78]. Should a permanent facility be developed, it is assumed that similar compensation, estimated at \$0.6/MWh, will be extended to nearby communities, as is done for existing waste storage locations [52].

To capture the effects of the uncertainty in social cost estimates, the sensitivity analysis considers scenarios where social cost parameters are 25 % lower and 25 % higher than the baseline values.

3.4.10. Exchange rate

The exchange rate used is 1218.36 KRW/USD, which is the recent five-year average from 2019 to 2023 [79].

Tables 2, 3, and 4 summarize key parameters used for selected technologies in this analysis in 2023 and 2035, respectively. Detailed parameter informati on for all technologies, including underlying data and assumptions, is available in Appendix C.

4. Results, sensitivity analysis, and discussion

4.1. LCOE comparison by technology

Figs. 1 and 2 represent LCOE without social costs. In Fig. 1, which presents LCOE by generator type for 2023 and 2035, "LCOE 2023" denotes the LCOE for each generator type in 2023, while "LCOE 2035" shows the Moderate LCOE scenario for 2035. Additionally, "LCOE 2035 (Low to High)" illustrates the range of LCOE values between the Conservative and Advanced scenarios for 2035.

Our LCOE analysis indicates that RE costs are expected to decrease by 28–41 % by 2035 from 2023 levels. In contrast, the LCOE for nuclear is projected to increase by 15 %, while coal is expected to decrease slightly, and natural gas is projected to decrease by 46 %. The main factor for decreasing RE cost is a decrease in OCC. For nuclear, the increase in OCC (2.55 %/y) drives the overall increase in that technology's LCOE. For coal and natural gas, while OCC increases, decreases in fuel costs drive decreases in LCOE. For natural gas, fuel costs are expected to decline 68 % by 2035.

Large-scale solar PV (20 MW and 100 MW) is projected to be the most cost-competitive energy source at \$47–48/MWh in 2030, while co-firing of GH and GA is the least cost-competitive at around \$260/MWh. LCOE for floating offshore WT, around \$175/MWh in 2023, is projected to decrease by more than a third by 2035 due to technological advancements and economies of scale.

While the range of LCOE for RE in 2035 across scenarios spans \$21–37/MWh, nuclear and coal show narrower ranges (less than \$1/MWh and \$11/MWh, respectively) due to less variation in OCC and stable fuel costs across scenarios. In contrast, the LCOE for natural gas in 2035 exhibits a wider range, \$54/MWh, due to greater variability in fuel costs (\$4.8–9.7/MMBtu) across scenarios.

Fig. 2 shows how LCOE is expected to change beyond 2035, with RE costs decreasing by 38–56 % by midcentury relative to 2023 levels. Solar PV of all sizes is projected to be the most cost-competitive energy source at \$28–36/MWh in 2050, and natural gas the least competitive. LCOE for offshore wind is projected to decrease to \$73–90/MWh by 2050 due to technological advancements and economies of scale. LCOE for nuclear increases by 44 %, mainly driven by an increase in OCC. Coal and natural gas is not shown after 2040, because those generators are assumed to be retrofitted as GH and GA co-firing generators. For natural gas, the LCOE decreases between 2023 and 2030 but rises again thereafter due to a decline in CF—from 33 % in 2030 to 18 % in 2040.

Notably, there are several crossover points for the costs of RE and

Table 2
Financial parameters for LCOE calculations.

Key parameters	PV 1 MW	Land-based WT	Fixed Offshore WT	Nuclear	Coal	Natural Gas
Inflation rate	2.0 %	2.0 %	2.0 %	2.0 %	2.0 %	2.0 %
Rate of return on equity (RORE)	8.0 %	8.0 %	8.0 %	8.0 %	8.0 %	8.0 %
Tax rate	19 %	19 %	19 %	19 %	19 %	19 %
Debt fraction	80 %	80 %	80 %	80 %	80 %	80 %
Nominal interest rate	4.4 %	4.4 %	4.4 %	3.8 %	3.8 %	3.8 %
Nominal WACC	4.5 %	4.5 %	4.5 %	4.0 %	4.0 %	4.0 %
Cost recovery period (years)	30	30	30	60	30	30
Nominal Capital Recovery Factor	6.1 %	6.1 %	6.1 %	4.5 %	5.8 %	5.8 %
Project finance factor	1.081	1.081	1.081	1.076	1.076	1.076
Construction finance factor	1.027	1.045	1.073	1.392	1.174	1.105

Table 3
Key parameters for LCOE calculations (2023).

		Key parameters	PV 1 MW	Land-based WT	Fixed Offshore WT	Nuclear	Coal	Natural Gas
Social LCOE	LCOE	CF/Calibrated CF (%)	13.8 %	20.0 %	29.4 %	83.4 %	53.8 %	41.7 %
		Annual energy production (kWh/kW)	1208	1756	2575	7308	4717	3650
		FCR	5.1 %	5.1 %	5.1 %	3.1 %	4.8 %	4.8 %
		CAPEX ($/kW)	$ 1194	$ 2144	$ 5351	$ 4247	$ 1809	$ 1275
		FOM ($/kW-yr)	$ 29	$ 46	$ 105	$ 139	$ 66	$ 42
		Fuel costs ($/MWh)	–	–	–	$ 5	$ 71	$ 212
		Compliance cost of carbon ($/MWh)	–	–	–	–	$ 0.8	$ 0.4
		Regulatory costs ($/MWh)	–	–	–	$ 2.5	$ 1.5	$ 1.2
		Potential carbon cost ($/MWh)	–	–	–	–	$ 7.4	$ 3.2
		Air pollution cost ($/MWh)	–	–	–	–	$ 10.3	$ 3.2
		Accident risk cost ($/MWh)	–	–	–	$ 7.5	–	–
		Subsidies for areas near high-level radioactive waste facilities ($/MWh)	–	–	–	$ 1.2	–	–

Table 4
Key parameters for LCOE calculations (2035).

		Key parameters	PV 1 MW	Land-based WT	Fixed Offshore WT	Nuclear	Coal	Natural Gas
Social LCOE	LCOE	CF/Calibrated CF (%)	14.6 %	21.2 %	32.4 %	83.2 %	40.2 %	21.9 %
		Annual energy production (kWh/kW)	1280	1856	2838	7284	3517	1916
		FCR	5.1 %	5.1 %	5.1 %	3.1 %	4.8 %	4.8 %
		CAPEX ($/kW)	$ 742	$ 1527	$ 3278	$ 5745	$ 2193	$ 1546
		FOM ($/kW-yr)	$ 20	$ 40	$ 85	$ 139	$ 62	$ 37
		Fuel costs ($/MWh)	–	–	–	$ 5	$ 41	$ 67
		Compliance cost of carbon ($/MWh)	–	–	–	–	$ 13.5	$ 5.9
		Regulatory costs ($/MWh)	–	–	–	$ 2.5	$ 1.5	$ 1.2
		Potential carbon cost ($/MWh)	–	–	–	–	$ 121.2	$ 52.7
		Air pollution cost ($/MWh)	–	–	–	–	$ 10.3	$ 3.2
		Accident risk cost ($/MWh)	–	–	–	$ 7.5	–	–
		Subsidies for areas near high-level radioactive waste facilities ($/MWh)	–	–	–	$ 1.2	–	

conventional resources. Nuclear has the lowest LCOE in 2023, but in the late 2020s, as OCC for solar PV declines while rising for nuclear, larger solar projects (20 MW and 100 MW) become the cheapest source of new power; and from 2035 onward, solar of all sizes becomes cheaper than nuclear. In 2023, coal's LCOE sits between those of land-based and fixed offshore wind, but the latter becomes cheaper than coal in 2030. From 2040, coal's LCOE exceeds that of floating offshore WT.

While these crossover points are informative for comparing generation-side costs, they should not be interpreted as representing the least-cost solution for the overall power system. The LCOE metric excludes integration costs such as transmission expansion, balancing, reserve provision, and curtailment management, all of which can alter the relative competitiveness of different technologies. For example, the rapid cost decline of solar PV implies earlier crossover with nuclear and coal; however, high PV penetration also requires additional flexibility resources to address variability and the day–night supply–demand mismatch. In addition, many renewable sites are distant from major load centers, necessitating significant transmission expansion and raising costs through grid reinforcement and congestion management. Similar challenges also apply to nuclear projects, which are typically sited far from demand regions and require dedicated high-capacity lines. Public acceptance further complicates both renewable and conventional projects, as large plants or new transmission corridors often face strong opposition. These factors underscore that the crossover points in Fig. 2 should be understood as indicators of generation-side competitiveness, not the system-optimal mix. Nevertheless, such comparisons remain valuable for illustrating long-term cost trajectories and highlighting when renewables begin to outcompete conventional resources on a pure generation-cost basis.

The results above are based on a single CF assumption for RE. To assess the impact of resource quality, we defined two alternative scenarios, Preferred and Ideal, each with higher CF assumptions. Fig. 3 presents the resulting LCOEs under these scenarios. For instance, if the CF for PV increases by 2.2 percentage points (%p), LCOE decreases by

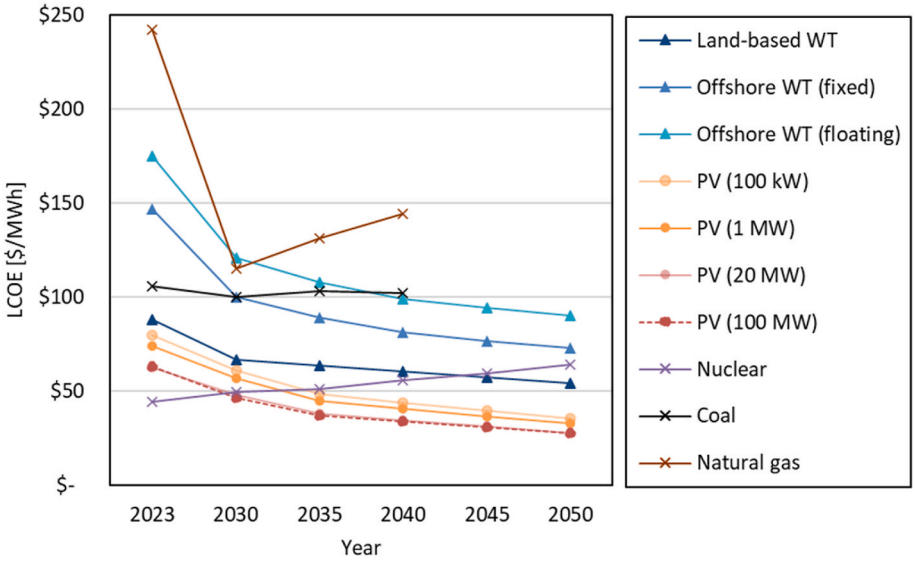


Fig. 2. LCOE projections for generation resources, 2023–2050.

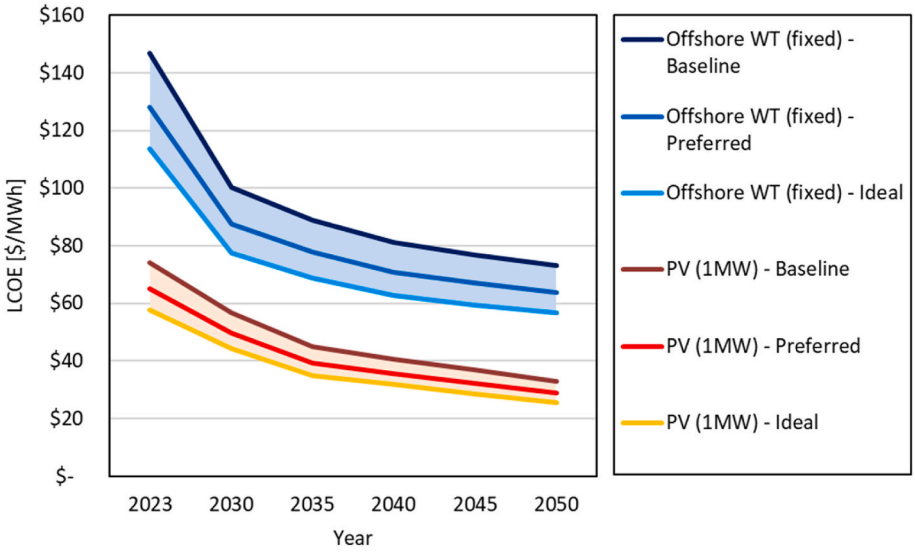


Fig. 3. LCOE projections for 1 MW PV and fixed offshore WT with varying CF.

approximately 12 %. In the Ideal scenario, a CF of 4.2 %p above Baseline yields an LCOE reduction of 22 %. A similar trend is observed for fixed offshore WT: increases in CF of 5.4 %p and 8.6 %p lead to LCOE reductions of 13 % and 23 %, respectively. These findings highlight the importance of identifying and developing project sites with higher resource potential. However, the potential for lower LCOE must be weighed against the possibility that high-CF sites may also entail higher OCC, which could offset the savings. While limited data availability restricted our ability to reflect this trade-off for PV, we provide a detailed analysis of this relationship for offshore WT in the following subsection.

4.2. Social LCOE

Fig. 4 compares the LCOE trends of renewables and conventional power sources, incorporating social costs of the latter. For nuclear, incorporating social costs (primarily accident risk costs) increases the LCOE by 14–20 %, making solar PV cheaper than nuclear sooner than in scenarios without social cost considerations. For 100 kW solar PV (the smallest and least economic project size), this shifts the cost crossover point with nuclear from the mid-2030s to the early 2030s; for larger solar PV projects (20 MW and 100 MW), the cost crossover with nuclear happens sooner. For coal and natural gas, incorporating social costs (primarily carbon costs) increases their LCOE significantly—by 113 % for coal and 41 % for natural gas in 2030, with coal becoming the most expensive energy source before then.

Fig. 5 compares the LCOE of fossil fuel generation with co-firing hydrogen-based green fuels in two cases: (a) excluding and (b) including social costs. The results show that co-firing green fuels is always more expensive than conventional fuels, regardless of whether social costs are incorporated. Higher LCOE figures for co-firing green fuels are mainly due to the high costs of those fuels. The price of GH, for example, is $86/MMBtu in 2023, projected to decrease to $50/MMBtu in 2035, and $44/MMBtu in 2050. Although the cost of GH decreases by roughly 49 % from 2023 to 2050, it remains expensive relative to other fuel cost projections (e.g., $6.5/MMBtu for LNG in 2050). Considering that shipping costs account for more than two-thirds of the total cost of imported GH, we also evaluate LCOE assuming domestically produced

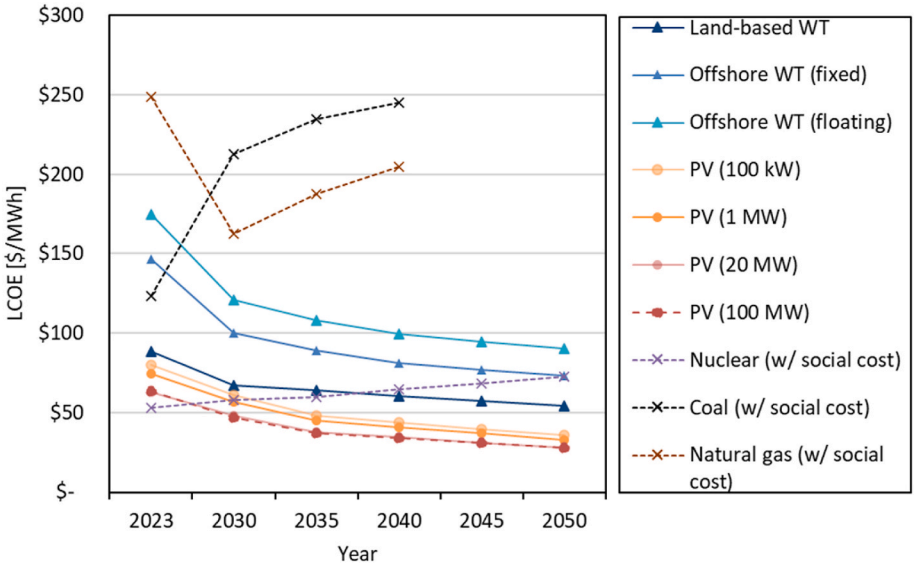


Fig. 4. Social LCOE for all generation resources, 2023–2050.

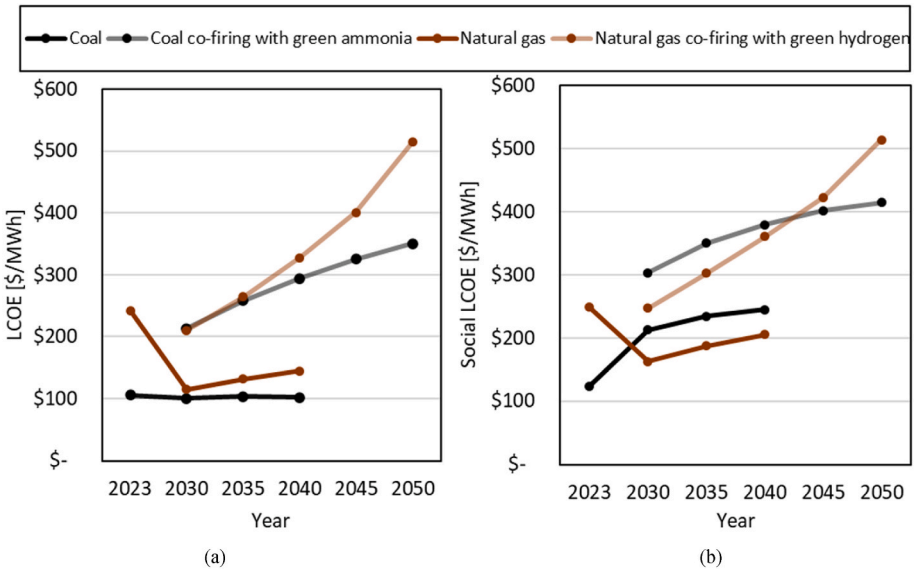


Fig. 5. LCOE and social LCOE for co-firing green fuels, 2023–2050.

GH at a lower cost of $17/MMBtu (equivalent to approximately $2/

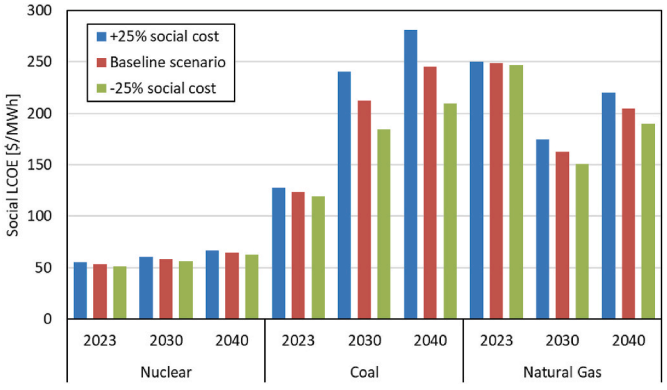


Fig. 6. Sensitivity of social LCOE to ±25 % changes in social cost assumptions for conventional technologies.

H_2kg) in the following subsection.

Fig. 6 illustrates how the social LCOE changes when the assumed social cost values are varied by ±25 %. For nuclear power, the social LCOE changes by 3–4 % from the baseline scenario across all years. In contrast, coal and natural gas show a greater sensitivity in later years compared to the present. For coal, the variation is 4 % in 2023 but increases to 13 % and 15 % in 2030 and 2040, respectively. Natural gas exhibits a smaller change, with a variation of 1 % in 2023 and 7 % in both 2030 and 2040. This higher sensitivity for coal is due to its greater share of social cost components, such as pollutants and carbon emissions, compared to other generation technologies.

4.3. Future LCOE for renewables and nuclear energy

Fig. 7 compares the future LCOE for RE with nuclear, which had the lowest LCOE in 2023. For 20 MW solar PV systems, under the Advanced scenario (which assumes a 53 % OCC reduction for PV by 2035), LCOE falls below that of nuclear by 2030. Smaller, 1 MW PV systems achieve cost parity with nuclear in 2030. Under the Conservative scenario,

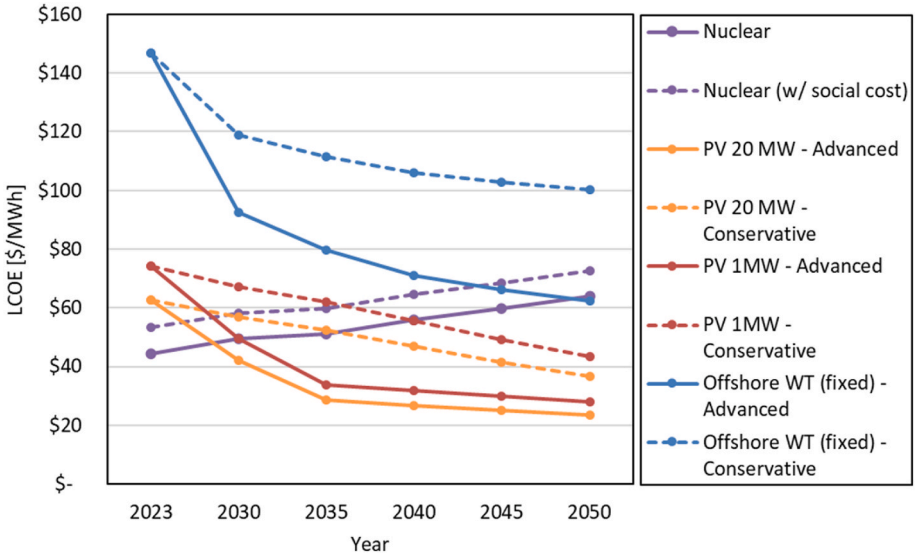


Fig. 7. Nuclear LCOE and social LCOE compared to LCOE for solar PV and fixed offshore WT, Conservative and Advanced Scenarios.

which assumes an 18 % OCC reduction for PV by 2035, 20 MW solar PV achieves cost parity with nuclear after 2035, while the LCOE of 1 MW PV falls below that of nuclear in 2040. Fixed offshore WT become cheaper than nuclear in 2050 under the Advanced scenario. These results indicate that limited technological innovation and constrained R&D could delay the LCOE crossover between PV and nuclear, prolonging society's exposure to the safety risks associated with nuclear power. Similarly, continued R&D and technological innovation may be necessary to achieve potential cost reductions in solar PV and the resulting cost savings for consumers.

4.4. LCOE comparison with other studies

While direct comparison with other Korean LCOE studies, such as Lee [52,52], is difficult due mainly to differences in data, assumptions, and methodology, the findings presented below provide important insights into the relative cost-competitiveness of different generation technologies in Korea.

Fig. 8 compares this study's LCOE results with those from Lee [52]. LCOE for RE technologies are 21–39 % lower in this study than in Ref. [52]. This is largely due to a difference in the cost recovery period, which is the period over which a generator's cost and performance are levelized [54]. This study sets the cost recovery period of RE technologies at 30 years, compared to 20 years in prior research [52]. According to research from NREL, typical module performance warranties have increased to 25–30 years, while the expected lifetime of utility-scale projects now exceeds 30 years [80]. Based on a survey of wind industry professionals, the average anticipated useful life of land-based wind power plants is estimated to be 30 years [81].

A 20-year payback period has long been standard in Korea, shaped

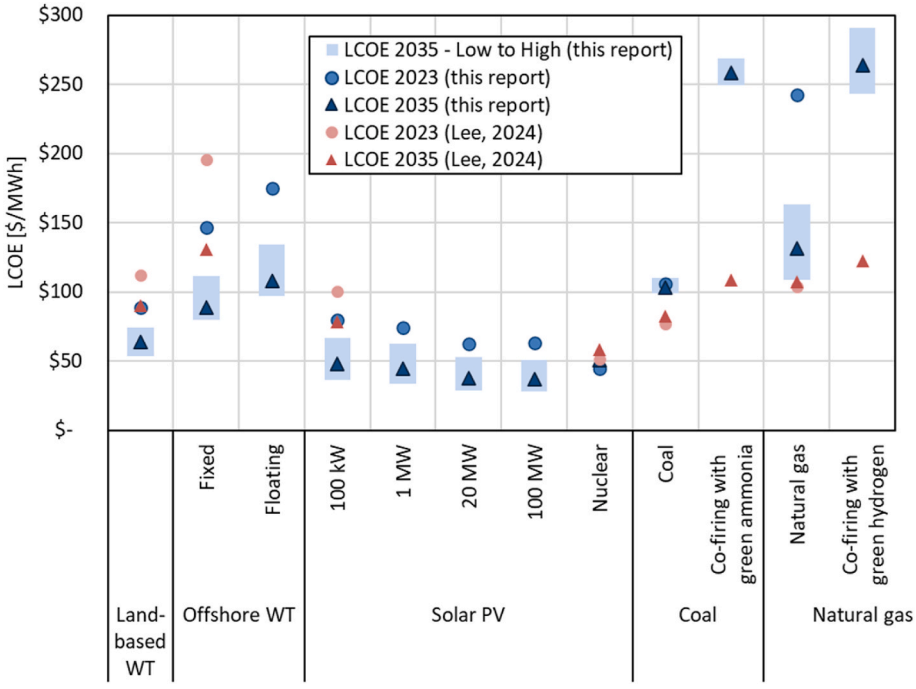


Fig. 8. LCOE for all generation resources compared with Lee [52].

by policy instruments such as the Feed-in Tariff and fixed-price contracts based on renewable energy certificates (REC), as well as associated insurance and financing practices. While this 20-year norm was fixed by REC-based contract regulations before 2017, recent revisions now allow more flexible contract durations depending on project characteristics [82]. Reflecting this evolving context—and in alignment with international trends, such as the 25- and 30-year project lifetimes used in IEA [42] and NREL [80] studies, respectively—this study adopts a 30-year payback period as the baseline, with 20- and 25-year scenarios explored through sensitivity analysis.

Fig. 9 presents results for a 1 MW PV system and a fixed offshore WT under shorter cost recovery periods. When the recovery period is reduced to 25 years, the LCOE increases by 9 % for both technologies, compared to a Baseline scenario; when further reduced to 20 years, the LCOE increases by 23 %. This is because shortening the cost recovery period increases the capital recovery factor from 4.7 % to 5.4 % and 6.4 % for 25-year and 20-year periods, respectively, which in turn raises the share of CAPEX in the LCOE. This highlights the significant impact of the cost recovery period on LCOE, as extending a generator's operational lifetime can substantially improve cost competitiveness by spreading capital costs over a longer period. Additionally, reflecting the recent trend of longer operating lives for RE technologies in longer cost recovery periods can contribute to lowering their LCOE.

For nuclear, the LCOE in this study and in Lee's study [52] show a similar trend, while coal and natural gas exhibit different trends, mainly due to assumptions around fuel costs and CF. This study assumes that fuel costs vary annually, whereas [52] assumed constant fuel costs. In this study, fuel costs start highest in 2023 and decrease gradually thereafter. Additionally, CF in this study varies in consideration of Korea's national average CF, while a constant value of 80 % is assumed in Ref. [52].

Furthermore, according to a recent BNEF report, the global benchmark LCOE is expected to decrease by 29 %, 24 %, and 15 % for 1 MW PV, land-based WT, and fixed offshore WT, respectively, by 2035 [83]. In this study, a similar trend is observed, though with slightly greater reductions of 35 % and 28 % for 1 MW PV and fixed offshore WT, respectively, and 18 % for land-based WT. However, the LCOE for RE in this study is 13–128 % higher than that reported by BNEF, indicating there is still potential for further cost reductions in line with the global trend.

4.5. Sensitivity analysis with varying carbon emission responsibility ratios

As mentioned previously, the CERR for power generation companies will increase from the current level of 10 % between 2026 and 2030 [73]. Fig. 10 compares the LCOE for coal and natural gas under various CERR with the LCOE for fixed and floating offshore WT. It shows that when the CERR increases from 10 % to 25 %, fixed offshore WT achieves cost parity with coal before 2030, while floating offshore WT becomes cheaper than coal in the early 2030s. At a 75 % or higher CERR, natural gas becomes cheaper than coal beginning in 2030. These results show that the CERR significantly impacts the LCOE for fossil-fueled generators, and that increasing this ratio improves the relative economics of less carbon-intensive generating technologies.

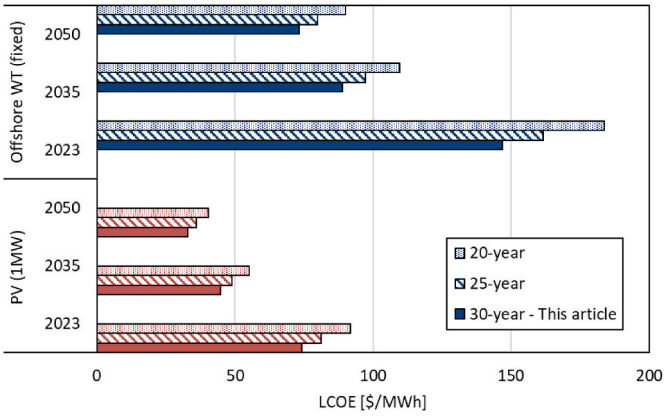


Fig. 9. LCOE of 1 MW PV and fixed offshore WT, assuming shorter cost recovery periods.

4.6. The potential of RE-ESS hybrid systems as balancing resources

GH is expected to serve two primary roles in future energy systems: enabling the continued use of natural gas power plants under net-zero scenarios and providing a zero-emission resource for system balancing. However, as shown in the previous results, co-firing GH with natural gas could be the most expensive option in the future due to the high cost of GH.

In this study, the assumed GH cost is based on imported hydrogen. Although the production cost is estimated at around \$2/H_2kg by 2030 [84] and around \$1/H_2kg by 2050 [85,85,86] in favorable countries (i.e., with rich RE resources), transportation and shipping costs can add \$3.1–3.8/H_2kg (e.g., imports from Australia [70]), which translates to around \$4.1–4.8/H_2kg even in 2050. If the cost of imported GH remains high, domestically produced GH could become an economically viable alternative. When electrolyzers operate for more than 6000 h per year, electricity price becomes the dominant factor in GH production cost, provided that operation can be sustained for over 70 % of the annual hours [87]. In such cases, electricity prices between \$20–40/MWh can produce GH at approximately \$2/H_2kg. The LCOE of PV in this study falls within this range after 2035, supporting the feasibility of domestic GH production in Korea. Furthermore, if GH can be produced using electricity from REs that would otherwise be curtailed due to oversupply, its production cost could be even lower. To capture the sensitivity, our analysis also considers optimistic cases with GH costs as low as \$1/H_2kg, serving as proxies for situations where curtailed or even negative-priced renewable electricity is available.

Fig. 11 presents the results of an LCOE comparison of flexibility resources, considering scenarios assuming a GH price of \$1–6/H_2kg with 100 % GH firing under two capacity factors: 18 % and 80 %. Under this assumption, the LCOE of a 100 % GH-firing plant is highly sensitive to the cost of GH, decreasing from \$595/MWh to \$173/MWh as the GH price falls from \$6/H_2kg to \$1/H_2kg in 2040. In a more optimistic scenario with a higher capacity factor of 80 %, the LCOE decreases further, as shown in the red area, reaching \$189/MWh and \$105/MWh in 2050 under GH prices of \$2/H_2kg and \$1/H_2kg, respectively.

The high LCOE of GH-based generation creates an opportunity for RE and energy storage system (ESS) hybrid plants to offer a more cost-effective alternative. For comparison, three RE-ESS hybrid plants consisting of 1 MW PV and a 1 MW/4 MWh battery is considered, assuming a 10-year lifespan for the ESS. This configuration is designed to store renewable electricity and dispatch it flexibly to enhance system reliability [88,89]. The OCC of the battery system is assumed to be \$689/kW and \$591/kW in 2040 and 2050, respectively. The LCOE of this hybrid system is estimated at \$149/MWh in 2040 and \$123/MWh in 2050, falling between the LCOE values of GH-based generation under optimistic CF and hydrogen cost assumptions. These results suggest that, when considering zero-emission flexible resources, RE-ESS hybrid systems are likely to be more cost-effective than GH-based generation under most scenarios, unless GH is available at very low prices (e.g., \$1–2/H_2kg) and high CFs can be achieved.

It is important to note, however, that the LCOE estimates for GH-based generation rely on the assumed cost of electricity used for hydrogen production. While PV LCOE is used as the baseline for our domestic production case, our analysis also incorporates sensitivity scenarios, including hydrogen prices as low as \$1/H_2kg, which highlights the potential competitiveness of GH-based generation. At the same time, we acknowledge that this approach does not capture the full range

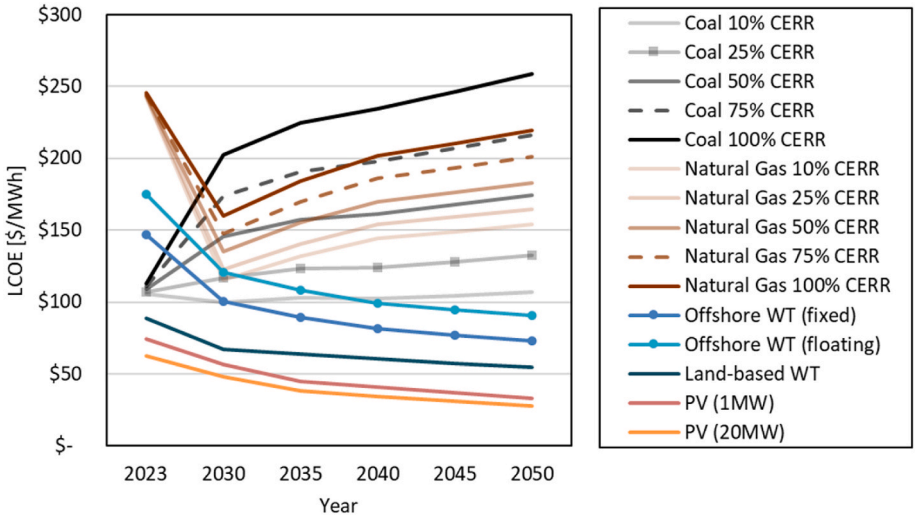


Fig. 10. LCOE comparison under various carbon emission responsibility ratios (CERR).

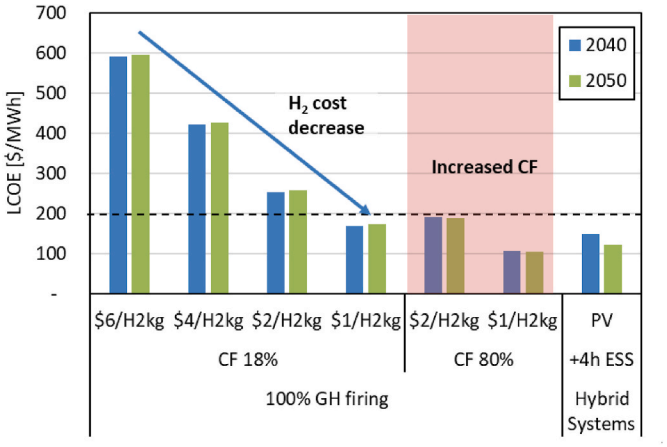


Fig. 11. LCOE comparison of flexibility resources.

of possible electrolyzer operating strategies: in practice, hydrogen costs may vary depending on utilization hours, reliance on grid-purchased electricity, or operation during non-solar hours to maximize capital utilization. These factors could influence the competitiveness of GH-based generation and merit further investigation in system-level analyses.

These results suggest that, when comparing zero-emission flexible resources, RE-ESS hybrid systems likely represent a more cost-effective option than GH-based generation unless GH is available at very low prices (e.g., $1–2/$H_2$kg). However, this comparison still has limitations in capturing the value that GH or natural gas-GH generators can offer—namely, firm capacity over prolonged periods, ensuring reliable power supply during multi-day extreme weather events or seasonal energy mismatches. As such, RE-ESS hybrid systems and GH- or natural gas-based technologies are likely to play complementary roles in future net-zero systems, rather than displacing each other. While LCOE comparisons provide a useful cost benchmark, future work should further evaluate the distinct system roles and potential synergies between short- and long-duration flexibility solutions.

4.7. LCOE of offshore wind turbines

Until this point, the LCOE of offshore WT was estimated using a single cost and a single CF assumption. However, for offshore WT, both the CF and the distance to shore can vary significantly depending on the installation location, leading to substantial differences in LCOE. To account for this variability, a more granular analysis is conducted for offshore WT.

Fig. 12 illustrates the key factors of LCOE for offshore WT across different offshore zones in Korea. Although nearshore areas generally exhibit lower mean CF compared to farther offshore regions as shown in Fig. 12(a), their proximity to the point of interconnection (POI) results in lower spur line costs. Furthermore, shallower water depths in these areas allow for the installation of fixed-bottom offshore WT as shown in the red area in Fig. 12(b), contributing to overall lower LCOE as shown in Fig. 12(c). Over time, as the OCC of offshore WT decreases by 45–47 % through 2050, the average LCOE decreases by 39 % as shown in Fig. 12(d).

Fig. 13 presents the distribution of Korea's offshore wind potential by LCOE at each POI. A higher concentration at lower LCOE values indicates more favorable economic potential. These results can serve as a basis for proactively planning grid infrastructure in regions with high resource quality. Specifically, areas such as Yeosu, Uido, and Taean exhibit strong potential for interconnection with offshore WT with LCOE below $90/MWh. However, in practice, offshore WT projects may still be prioritized near major demand centers—such as those connected to Incheon or Taean—even if their LCOEs are relatively higher. This reflects the strategic importance of WT proximity to high-load urban areas, which can offset the downside of higher generation costs.

5. Conclusions

LCOE is a standardized metric used to evaluate and compare the economic viability of diverse energy generation technologies in electric system planning and operation. It distributes the costs of expected generation from an energy resource over its expected useful lifetime using an assumed annual average CF. It also integrates key cost and performance parameters (e.g., capital expenditures, operational expenses, and CF) into a single value. This makes LCOE particularly useful for comparing technologies with differing capital and fuel requirements. Additionally, LCOE serves as an effective tool for exploring future projections of technological advancements, as it reflects the combined impact of these primary cost drivers and is relatively straightforward to calculate.

Our analysis shows that RE technologies—particularly solar PV and offshore WT—are expected to become the most cost-competitive generation options for Korea. By 2035, RE costs are projected to decline by 28–41 % relative to 2023 levels. During the same period, the LCOE for nuclear is expected to rise by 15 %, while coal will slightly decrease and

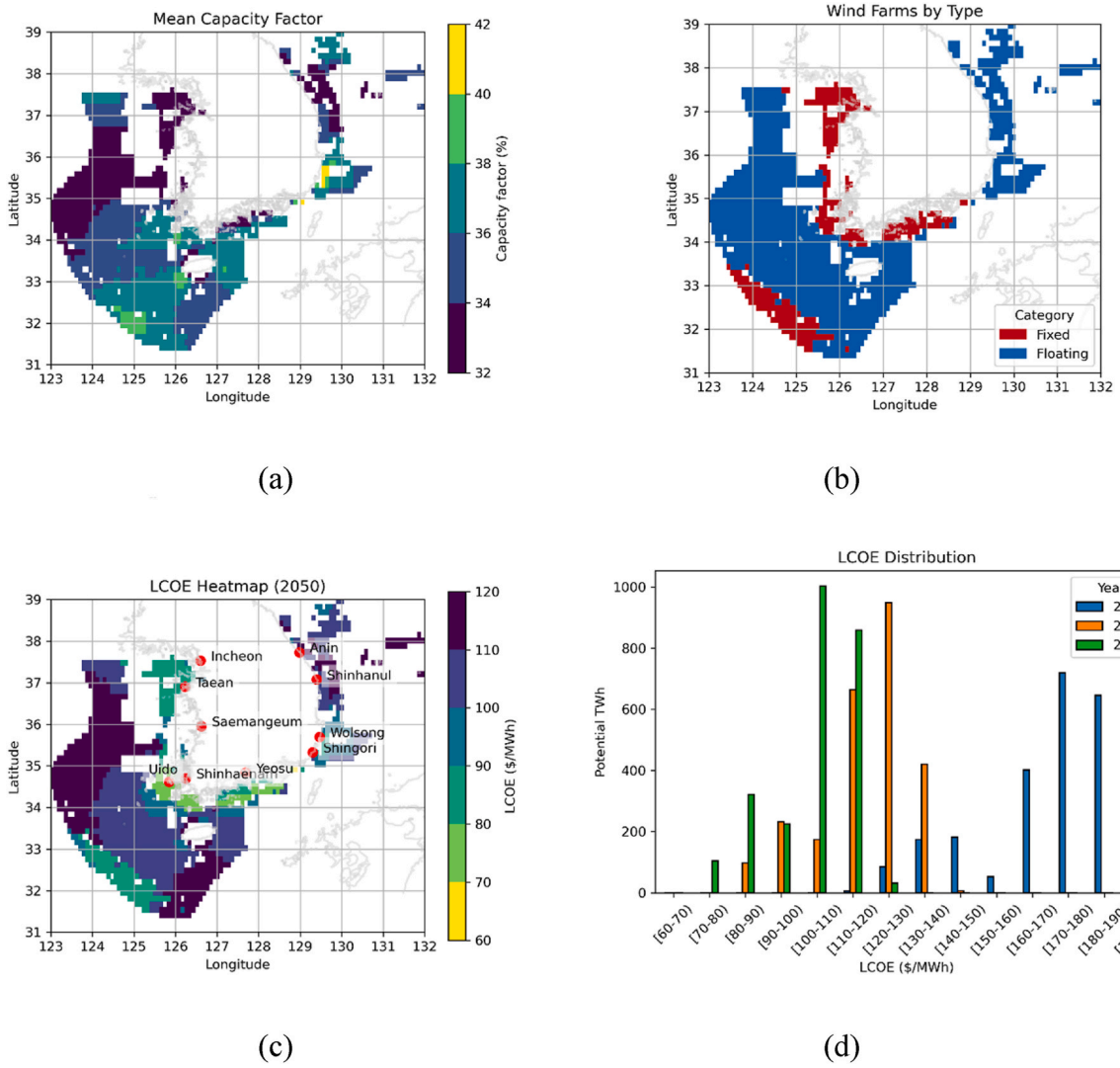


Fig. 12. Distribution of LCOE for Korean offshore WT: (a) Average CF; (b) Wind farms by type (fixed vs. floating); (c) LCOE in 2050; (d) LCOE distribution by year. Red dots in (c) represent potential points of interconnection (POI).

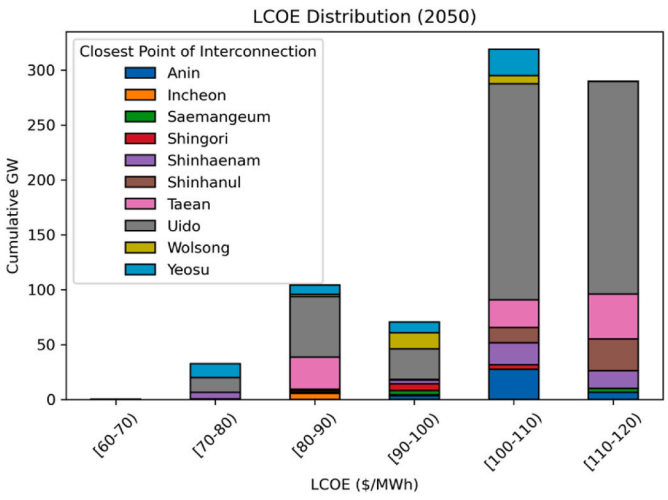


Fig. 13. LCOE distribution of offshore WT by the closest POI in 2050.

natural gas will decline by 46 % from 2023's elevated level. Larger solar PV systems (20 MW and 100 MW) are expected to become Korea's lowest-cost energy resource before 2030. The cost of RE is expected to decline further—by 38–56 % by 2050 relative to 2023—making it more cost-competitive than other resources even before social costs are accounted for. When social costs are included, PV is projected to be the most cost-competitive option from the early 2030s. As RE installation costs decline, multiple cost crossovers are projected: large PV will overtake nuclear before 2030, fixed offshore wind will intersect coal's LCOE by 2030, and floating offshore wind will reach parity with coal by the late 2030s.

Throughout most of the 2023–2040 period, natural gas is expected to remain the highest-cost resource, while GH co-firing with natural gas is projected to become the highest-cost resource between 2040 and 2050 if social costs are excluded. These trends are primarily driven by high fuel prices and declining CFs. However, at a CERR of 75 % or higher, natural gas becomes more economical than coal from 2030, highlighting the potential role of carbon policies in shifting relative cost-competitiveness. The high cost of GH-based generation also creates an opportunity for RE-ESS hybrids to serve as cost-effective, zero-emission flexible resources. Additionally, location-based LCOE analysis helps identify priority offshore wind project sites. Areas closer to shore, despite lower CFs, may offer lower LCOE due to reduced transmission

costs, though further refinement is needed to address grid availability and social acceptability, including navigational and fishing constraints.

Furthermore, the LCOE crossovers in this study should be interpreted as indicators of generation-side competitiveness, not as the least-cost system configuration. Integration costs—including transmission expansion, flexibility provision, and balancing—as well as non-economic barriers such as land constraints and public acceptance can shift these crossover points. These considerations highlight that the importance of system-level planning for identifying the optimal technology mix. Nevertheless, LCOE remains a valuable standardized metric for tracking long-term cost trajectories on the generation-side and signaling when renewables begin to outcompete conventional resources. At the same time, prioritizing cost-competitive generation investments in the power sector requires optimization modeling such as capacity expansion planning and hourly dispatch models that consider physical constraints on the power system [1,90].

In addition, to capture the marginal cost implications of integrating new generation into the grid, the concept of "system LCOE" can be considered. Since system LCOE depends on the configuration of the future grid—including its generation mix, flexibility constraints, and transmission bottlenecks—it is essential that future studies incorporate a range of plausible power system scenarios and evaluate system LCOE under these conditions.

These findings suggest that Korea can build a low-cost, low-carbon power system by accelerating the deployment of RE technologies, particularly solar PV and offshore wind, and strategically siting them in high-CF regions and/or areas with lower connection costs. Given the cost advantage of RE-ESS hybrid systems, policy should support their deployment as a dispatchable clean energy solution. At the same time, despite their higher costs, GH-based generators can play a complementary role by providing sustained, long-duration capacity that addresses the limitations of battery storage. A balanced strategy that recognizes the distinct value of each technology will be critical to achieving a reliable and cost-effective zero-carbon power system. These complementary dynamics should be further explored in future studies using system-level assessment frameworks such as system LCOE, which can capture the broader operational and reliability value of each technology.

Additionally, LCOE does not fully reflect non-technical risks that may significantly affect project cost and viability—such as public opposition to siting, permitting delays, supply chain disruptions due to geopolitical tensions, or long-term fuel price volatility. While these risks are difficult to quantify uniformly across technologies, they can indirectly increase LCOE by delaying construction schedules, escalating fuel and capital expenditures, or raising financing costs. Future modeling efforts may consider scenario-based or risk-adjusted frameworks to more comprehensively account for such uncertainties.

The sensitivity analysis on social costs underscores that social LCOE is influenced not only by the inclusion of externalities but also by uncertainty in how they are valued. Coal's relatively high sensitivity reflects its greater dependence on external cost assumptions, suggesting that emission-intensive technologies are more vulnerable to changes in valuation for environmental impacts or policy direction.

While this study incorporates social costs such as air pollution damages, carbon emissions, and nuclear accident risks, these values are treated in a deterministic manner and assumed to remain constant over time. However, such costs can vary significantly depending on geographic factors such as the location of fossil fuel power plants, population density, and local environmental conditions, as well as probabilistic aspects, particularly in the case of nuclear accident risks. For example, the extent of air pollution damages may depend on pollution control technologies or prevailing wind patterns, while accident-related costs may vary with event probabilities and socio-economic exposure. Future research should adopt spatially resolved and probabilistic approaches to better capture the uncertainty and distributional effects of social costs over time.

We recommend that Korea's government explore expanding carbon pricing mechanisms and strengthening emissions reduction targets to improve the cost-competitiveness of low-carbon resources. These actions could enhance energy security, reduce long-term electricity costs for consumers, and contribute to macroeconomic stability while supporting progress toward carbon reduction goals and grid resilience. Additionally, enabling grid flexibility, streamlining permitting processes, and investing in domestic GH production and advanced storage technologies will support a more resilient, affordable, and decarbonized energy future.

CRediT authorship contribution statement

Hee Seung Moon: Writing—original draft, Visualization, Formal analysis, Methodology, Data curation, Conceptualization. Sunhee Baik: Writing—review & editing, Formal analysis, Methodology, Conceptualization. Won Young Park: Writing—review & editing, Conceptualization, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Levelized Cost Of Energy Methodology in ATB

$$LCOE_{ATB} = \frac{FCR \times CAPEX + FOM}{CF \times 8,760\ hours/year} + VOM + Fuel - PTC \quad \text{Eq. (A.1)}$$

Table A.1
Definitions of variables used for fixed charge rate (FCR) calculation

Variable	Definition	Equation
Fixed Charge Rate (FCR)	Amount of revenue per dollar of investment that must be collected annually from customers to pay the carrying charges on that investment	$\boldsymbol{CRF \times PFF}$
Capital Recovery Factor (CRF)	Ratio of a constant annuity to the present value of receiving that annuity for a given length of time	$WACC \times \frac{1}{1-\frac{1}{(1+WACC)^t}}$
Weighted Average Cost of Capital (WACC)	Average expected rate paid to finance assets	$\frac{[1+(1-DF)((1+RROE)(1+i)-1)+DF((1+IR)(1+i)-1)(1-TR)]}{1+i}-1$
Debt fraction (DF)	Fraction of capital financed with debt	Varies by technology
Rate of Return on Equity (RORE)	Assumed rate of return on the share of assets financed with equity	Varies by technology
Real Interest Rate (IR)	Assumed interest rate on debt	$\frac{1+d}{1+i}-1$
Tax rate (TR)	Marginal state and federal tax before application of available federal tax credits	25.7 %
Project Finance Factor (PFF)	Financial multiplier to account for any applicable differences in depreciation schedule and tax policies; the investment tax credit (ITC) appears first to reduce the depreciable basis and a second time to apply the value of the credit (Ho et al., 2021)	$\frac{1-TR \times PVD \times \left(1-\frac{ITC}{2}\right)-ITC}{1-TR}$
Present Value of Depreciation (PVD)	Present value of depreciation	$\sum MACRS\ FD \frac{1}{[(1+WACC)\times(1+y)]^y}$
Depreciation fraction (FD)	Depreciation fraction, the fraction of capital depreciated in each year	Varies by technology
Depreciation period (MACRS)	Number of years in modified accelerated cost recovery system (MACRS) depreciation schedule	
Investment Tax Credit (ITC)	Tax credit received on the CAPEX of the system. ITC is not considered in this study.	

Source [54].

Table A.2
Definitions of variables used for capital expenditures (CAPEX) calculation

Variable	Definition	Equation
Capital Expenditures (CAPEX)	Expenditures required to achieve commercial operation of the generation plant ($/kW)	$ConFinFactor \times (OCC + GCC)$
Overnight Capital Cost (OCC)	Capital expenditures excluding construction period financing but including on-site electrical equipment (e.g., switchyard), a nominal-distance spur line (<1 mi), and necessary upgrades at a transmission substation	Varies by technology ($/kW)
Grid Connection Cost (GCC)[a]	Distance-based costs of spur lines over land, offshore wind plant export cable costs, and construction period transit costs	
Construction Finance Factor (ConFinFactor)	Portion of all-in capital cost associated with construction period financing	$\sum_{y=0}^{C-1} FC_y \times AI_y$
Accumulated Interest (AI)	Portion of the construction finance factor due to interest on a construction loan	$AI_y = 1+(1-TR)\times[(1+IDC)(year+0.5)-1]$
Capital fraction (FC)	Fraction of capital spent in each year of construction	Varies by technology

[a] In the original ATB, GCC only applies to offshore wind ($/kW). In this study, OCC is considered to already include GCC.
Source [54].

Table A3
Definitions of variables used for variables other than FCR and CAPEX calculation

Variable	Definition	Equation
Fixed Operation and Maintenance expenses (FOM)	Annual expenditures to operate and maintain equipment that are not incurred on a per-unit-energy basis ($/kW-yr)	
Net Capacity Factor (CF)	The ratio of actual annual output to output at rated capacity for an entire year (%)	
Variable Operation and Maintenance expenses (VOM)	Operation and maintenance costs incurred on a per-unit-energy basis ($/MWh)	
Fuel	Fuel costs to run a generating plant ($/MWh)	$Heat\ rate\ (MMBtu/MWh) \times Fuel\ costs\ (\$/MMBtu)$
Production Tax Credit (PTC)	Production tax credit value received by the project, normalized by the capital recovery period ($/MWh). PTC is not considered in this study.	$\frac{PTC_{full}}{1-TR} \times \frac{CRF}{CRF_{10\ years}}$

Source [54].

Appendix B. Baseline Year Costs

Table B.1
Reference of key cost parameters for the baseline year (2023) by generation resource

Resource		Description
Land-based Wind		OCC: Weighted average of investment costs from 73 land-based wind projects between 2019 and 2024 [57]. FOM: Used OPEX for land-based wind in 2023 from Lee and Lim [53,53].
Fixed Offshore Wind		OCC and FOM are derived from those of fixed offshore wind from Lee and Lim [53,53].
Floating Offshore Wind		OCC: Based on the OCC for fixed offshore wind from Lee and Lim [53,53], estimated using the OCC ratio between fixed offshore wind and floating offshore wind in the ATB [54]. FOM: Used OPEX for fixed offshore wind in 2023 from Lee and Lim [53,53].
PV	<100 MW	OCC and FOM are derived from those of PV in each class from Lee and Lim [53,53].
	≥100 MW	OCC: Weighted average of investment costs from two PV projects that applied for EBP between 2019 and 2024 ([57]. FOM: Used FOM for 20 MW PV from Lee and Lim [53,53].
Nuclear		OCC and FOM for nuclear are derived from Lee [52,52], heat rate from Ref. [52].
Coal		OCC: Weighted average of investment costs from three coal generators built between 2020 and 2023 [57], and from four coal generators included in Lee et al. [56,56]. FOM: Used OPEX for coal from Lee [52,52]. Heat rate: Used national average heat rate in 2023 [72].
Natural Gas		OCC: Weighted average of investment costs from four natural gas generators built between 2018 and 2024 [57], and from five natural gas generators included in Lee et al. [56,56]. FOM: Used OPEX for natural gas from Lee [52,52]. Heat rate: Used national average heat rate in 2023 [72].

Appendix C. Projection of Future Costs

Coal

IEA's World Energy Outlook (WEO) projects future fuel prices by region, with Japan being one such region [42]. Since WEO does not project future fuel prices for Korea, these have been estimated based on Japan's figures. Figure C.1 illustrates a linear regression of combined data points representing historical coal prices in Japan [92] and Korea [68] for the same years. A high R-squared value, 0.89, confirms a significant correlation between the two.

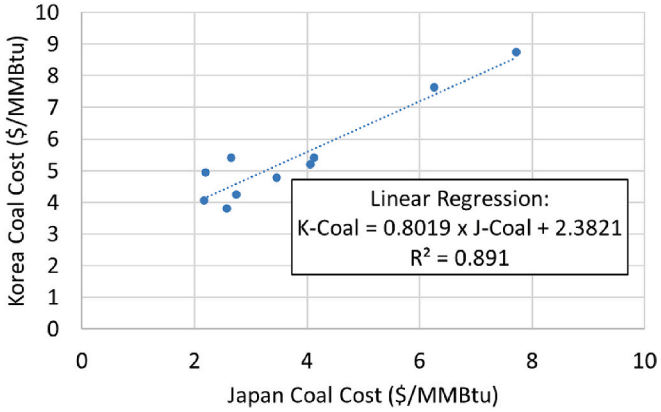


Fig. C.1. Comparison of per-unit coal costs in Korea [68] and Japan [92]

Based on the linear regression, Figure C.2 shows a comparison of the two countries' actual historical coal price trends, along with an estimate of Korea's based on Japan's. This demonstrates that the estimated prices closely follow Korea's actual historical prices in terms of major trends.

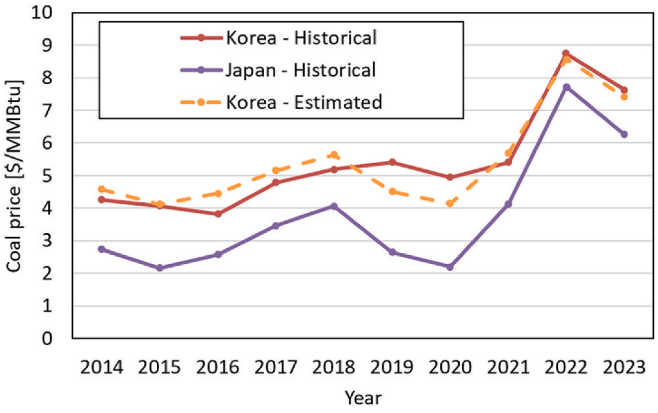


Fig. C.2. Historical coal price trend in Korea and Japan, and Korea's estimated coal costs based on Japan's

Figure C.3 displays Japan's historical coal prices alongside price scenarios presented in WEO [42]. Each price scenario accounts for changes in fuel demand in accordance with country-specific decarbonization scenarios.

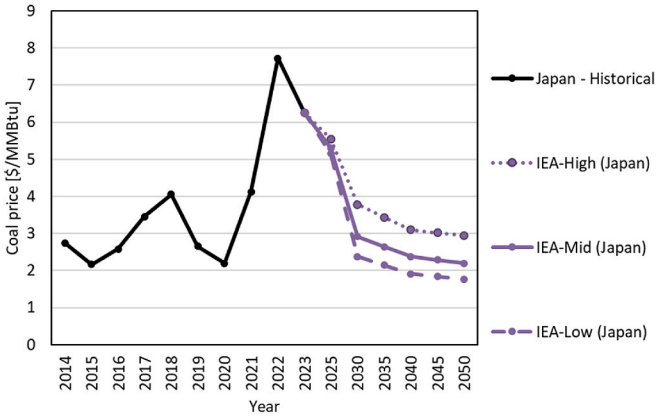


Fig. C.3. Coal cost projection for Japan [42]

Based on Japan's future coal cost, Korea's future coal cost is estimated as shown in Figure C.4. It indicates that coal prices will rapidly decrease to around $5/MMBtu in 2030, followed by a slower rate of decline to around $4/MMBtu.

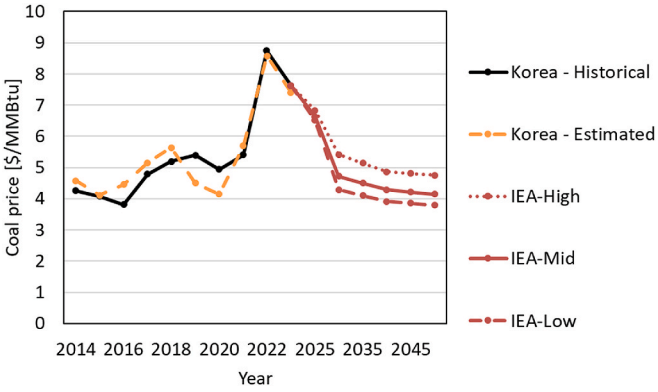


Fig. C.4. Coal cost projection for Korea (estimated based on [42])

Natural Gas

The approach used for coal was also used for natural gas. Figure C.5 illustrates a linear regression of the combined data points representing liquefied natural gas (LNG) prices in Japan and Korea for the same years. Although the R-squared value, 0.79, is lower than that for coal, it still confirms a high correlation between the LNG prices of the two countries.

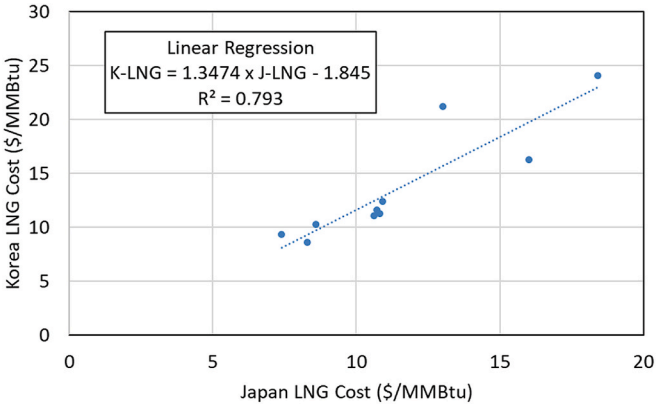


Fig. C.5. Comparison of per-unit LNG costs in Korea [68] and Japan [93]

By combining the relationship between LNG costs in Japan and Korea with projections of LNG costs in Japan, Korea's LNG costs can be illustrated as shown in Figure C.6. Starting from a level of $22/MMBtu in 2023, a sharp decline is expected, reaching a level of $5–10/MMBtu by 2030. Post-2030, prices are projected to stabilize across all scenarios.

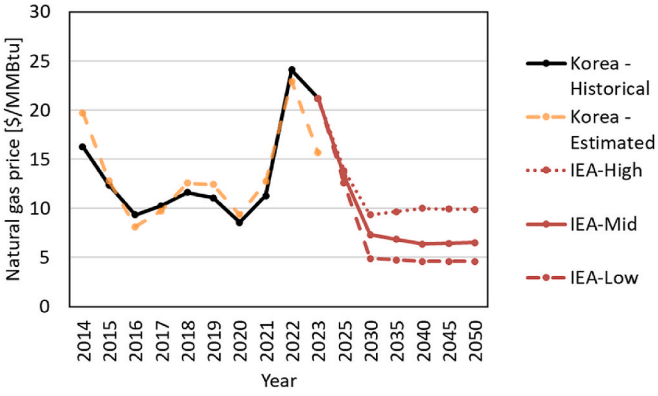


Fig. C.6. Korea LNG cost projection

Nuclear

According to ATB 2024 [94], the CAPEX of a nuclear power plant is structured as shown in Figure C.7. Around two-thirds of the total cost comes from direct costs, followed by indirect service costs, preconstruction costs, and supplementary costs. The largest portion, direct costs, primarily includes expenses for physical systems such as the reactor system, energy conversion system, and electrical equipment. Supplementary costs include insurance expenses, which account for 1.56 % of the total CAPEX.

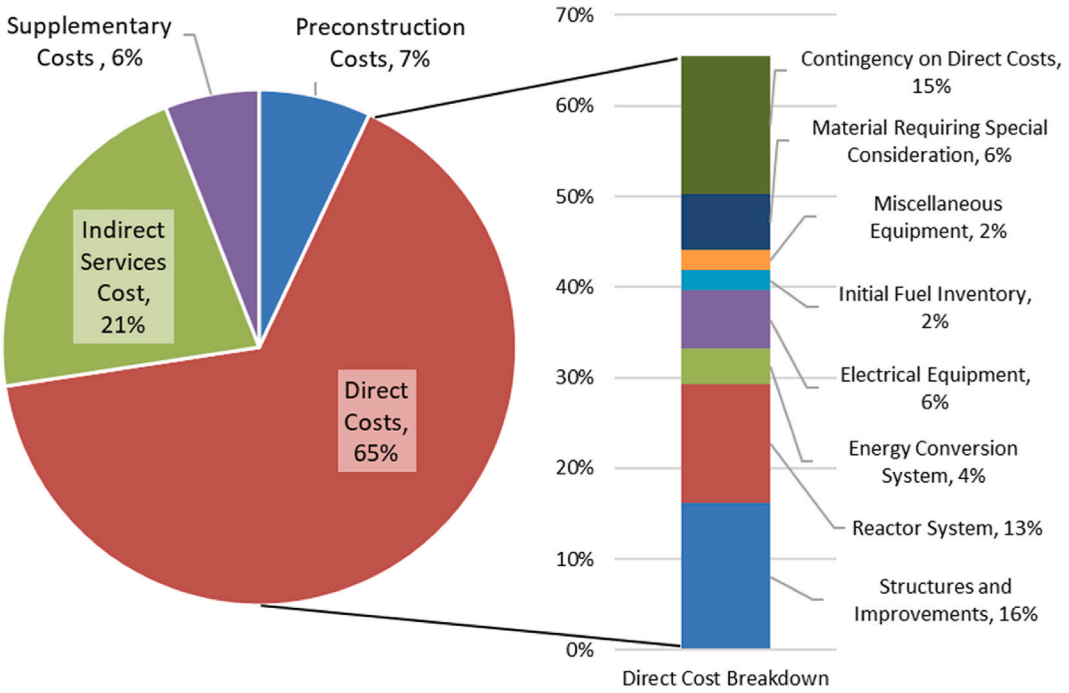


Fig. C.7. Cost breakdown of capital expenditures for a nuclear power plant [94]

When calculating LCOE for nuclear power plants, both direct costs (including capital, O&M, and fuel costs) and indirect costs (including those arising from routine plant operations) are typically considered. However, certain costs—such as those related to accident risk management, additional safety measures, policy compliance, and back-end processes (e.g., the subsequent nuclear fuel cycle)—remain controversial. These costs are sometimes categorized as direct costs, but are often excluded from analysis as external cost items.

In a recent study, the cost of nuclear accident risk management in Korea was estimated to be $7.50/MWh, based on the costs of the Fukushima nuclear accident (nearly 24 trillion yen) [52]. This figure includes damages compensation, decontamination, interim waste storage, and decommissioning. These costs were then adjusted to reflect Korea's income levels and population density.

In the ATB 2024 [54], cost projections are derived from the EIA's Annual Energy Outlook 2023 [69]). Detailed explanation is limited, and insurance costs do not appear to be explicitly included in the model. To account for these, the fixed O&M (FOM) cost was increased by 20 %. However, this adjustment also incorporates other costs, so it cannot be fully attributed to insurance. As ATB states: "Because the EIA fixed O&M costs for new power plants do not include property tax or insurance costs as part of their reported costs, we increase the fixed O&M by 20 %, which is approximately consistent with the internal property tax and insurance accounting that occurs within EIA's capacity expansion model."

In ATB 2024 [94], costs are categorized based on the Generalized Nuclear Code of Accounts. This approach combines historically observed costs with detailed estimates of modern designs to provide a representative cost range for nuclear energy [95]. Insurance costs included in CAPEX (Code 52) are defined as "insurance costs associated with the permanent plant to be capitalized with the plant." Based on the 2024 Conservative Scenario's CAPEX values, these costs (in 2030 USD) are estimated at $11,836 × 0.0156 = $184.64/kW for large reactors and $12,681 × 0.0070 = $88.77/kW for small modular reactors (SMRs). Other related costs are also identified, including construction insurance (Code 319); insurance for work-related injuries and accidents during construction; salary-related expenses, including taxes, insurance, fringe benefits, and other associated costs (Code 718); and taxes and insurance covering property taxes and insurance costs but excluding salary-related expenses (Code 76). Specific values for these cost components are not provided.

The costs outlined in ATB 2024 remain significant limitations. The report acknowledges that "total generating costs presented do not represent the full costs of operations, as they do not include market and operational risk management, property taxes, depreciation, interest costs, spent fuel storage costs, or returns on investment—key factors in decision-making about continued operation of a nuclear plant." To address this gap, property taxes and deactivation and decommissioning costs were introduced in this analysis as non-fuel O&M costs for new nuclear plants, estimating $1.23/MWh for levelized decommissioning and $1/MWh for the spent fuel disposal fee.

However, liability costs remain unaddressed in the ATB, and thus require further analysis. U.S. nuclear plants must carry $450 million in liability coverage, which is far below a worst-case scenario but exceeds the average cost of accidents. The World Nuclear Association suggests that "internalizing all costs from an accident … with further obligatory catastrophic accident coverage of $10–15 billion could be provided for a cost of 0.1–0.2 cents/kWh." In Korea, the liability coverage is stipulated by Nuclear Damage Compensation Act as 9 billion Special Drawing Rights (equivalent to $1.2B), which also has a limitation in covering the worst-case scenario. Since ATB does not consider safety and accident risks, we referenced [52], containing information on Fukushima, to incorporate these costs into our social LCOE calculation.

SMR

SMRs are an emerging technology for nuclear reactors with a power capacity of up to 300 MW per unit, or roughly one-third the generating capacity of traditional nuclear plants [96]. Nine countries are developing water-cooled SMRs for land-based power generation purposes [97]. In Korea, a plant using "System-integrated Modular Advanced Reactor" technology, capable of generating 100 MW, received standard design approval in 2024, while the "i-SMR" project, with a capacity of 170 MW, is already under development. The i-SMR project aims to commercialize SMRs in the

2030s, while Korea Hydro and Nuclear Power (KHNP) has set a cost target of less than $3500/kW Korea Hydro & Nuclear Power [98]. This target falls in the lower range of other SMR estimates: $2850–7700/kW [99], €3160–20,153/kW [100], and $3985–4844/kW [101]. Due to limited access to detailed SMR cost data, LCOE analysis of SMR is not included in this study.

Renewables

According to IRENA's analysis of historical LCOE trends, solar PV's LCOE in Korea shows a sharp decline in the early 2010s, mirroring past experience in the U.S. LCOE for land-based WT in Korea exhibits an overall decrease from 2010 to 2022, albeit with fluctuations The International Renewable Energy Agency [102].

For generators other than offshore WT, future OCC trends are derived from the NREL ATB [54]. Korean land-based wind OCCs align closely with ATB Class 10. As offshore wind is still emerging in Korea, its costs are assumed to align with ATB projections by 2040, subsequently following the same rate of change as in ATB scenarios (Class 7 for fixed WT, Class 14 for floating WT). OCC projections for wind technologies in Korea are shown in Figure C.8.

Utilizing the wind deployment targets outlined in Korea's 11th Basic Plan, the projected costs under three scenarios correspond to learning rates of 6–13 % for land-based wind and 8–17 % for offshore wind. We assume that WT are deployed following a linear regression trend based on the annual targets specified in the Plan. For example, a learning rate of 8 % means that for every doubling of cumulative installed capacity, the cost decreases by 8 %. Therefore, the faster cumulative capacity grows, the more frequently it doubles, leading to faster cost reductions due to learning effects (Figure C.9).

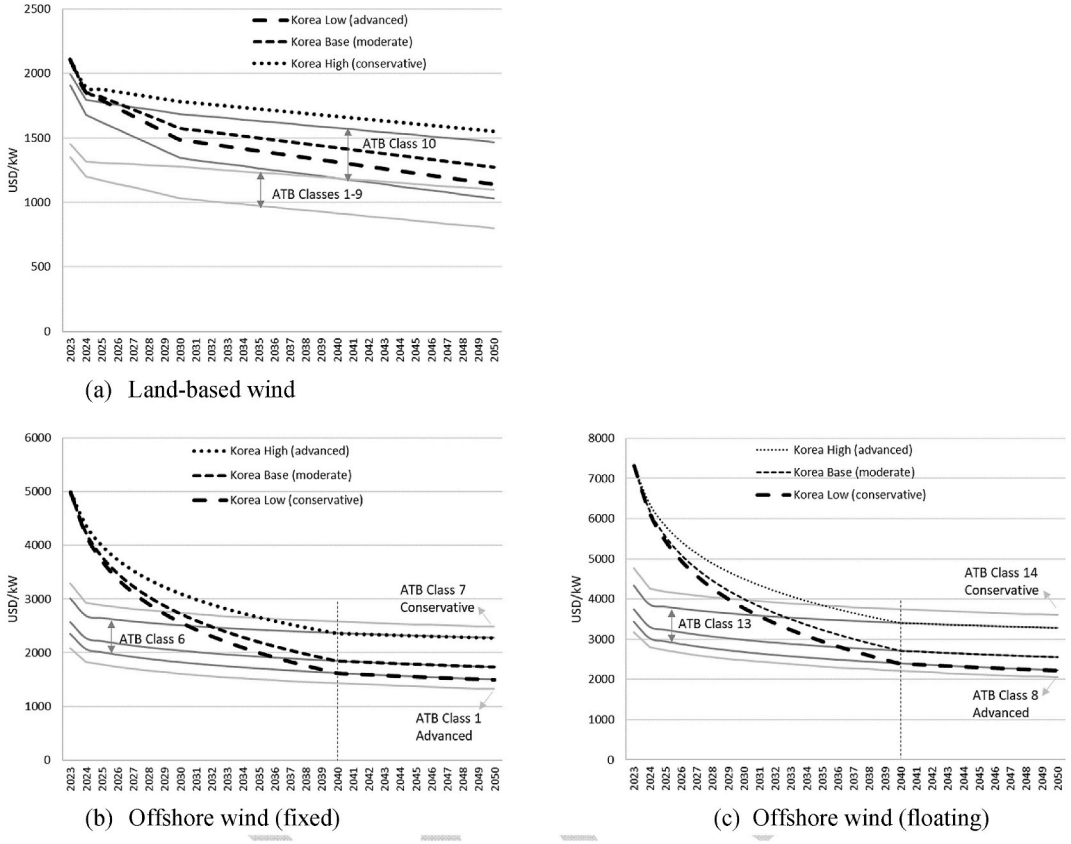


Fig. C.8. OCC projections for wind technologies in Korea

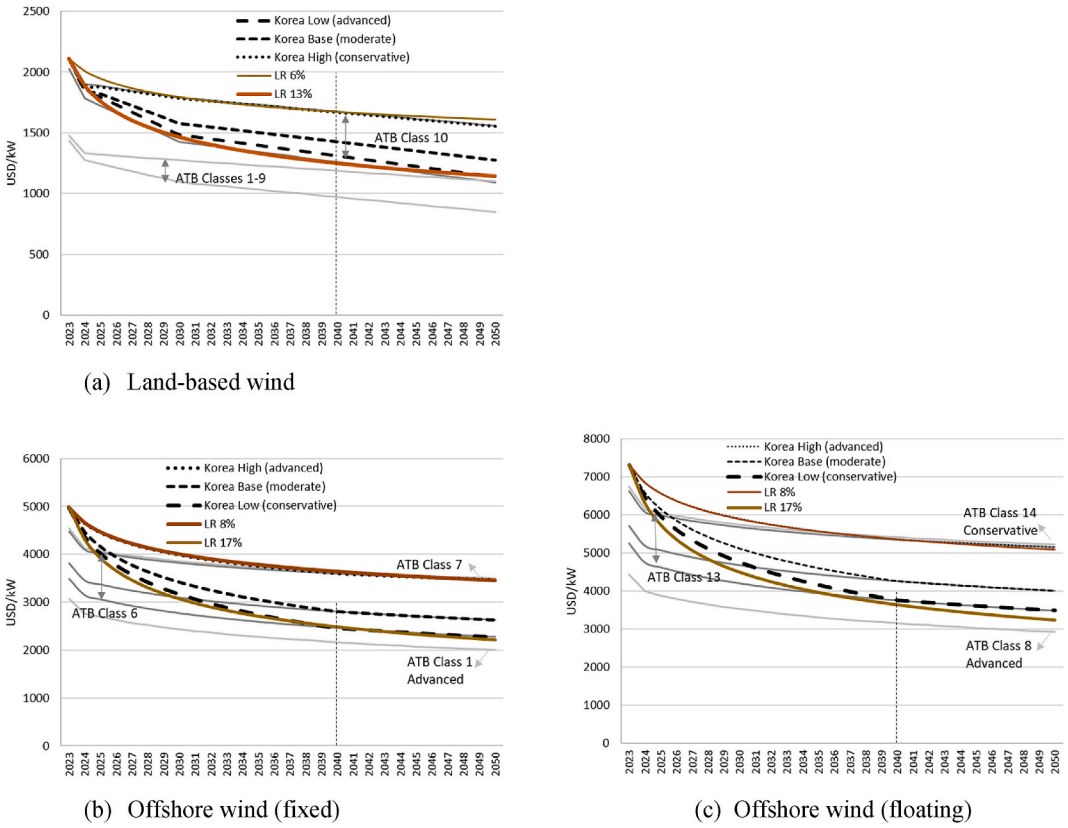


Fig. C.9. OCC projections for wind technologies in Korea with corresponding learning rates

Co-firing of green ammonia and green hydrogen

In Korea's 11th Basic Plan, which covers the period 2024–2038, the proportion of power generation from "clean hydrogen" and "clean ammonia" has been set as 2.4 % by 2030 and 6.2 % by 2038 [4]. "Clean" hydrogen and ammonia include not only "green" hydrogen and ammonia, which are produced using renewable energy (RE), but also "blue" hydrogen and ammonia. "Blue hydrogen" refers to hydrogen that is produced by reforming fossil fuels such as natural gas and ammonia, where the carbon dioxide emitted during the process is captured and removed. In this study, we only assumed green hydrogen and ammonia are used for co-firing.

Korea's 11th Basic Plan also references the opportunity to co-fire natural gas generation with 50 % hydrogen and co-fire coal generation with 20 % ammonia [4]. Accordingly, for GH, a co-firing rate of 50 % is assumed for 2030, and a 100 % switch to green hydrogen then occurs by 2050, in alignment with Korea's net zero target [103]. For GA, a co-firing rate of 20 % is assumed for 2030, increasing to 60 % by 2050. Because co-firing rates above 60 % for ammonia are known to reduce generation efficiency, a maximum co-firing rate of 60 % is assumed for 2050 [104]. Co-firing rates of GH and natural gas are based on volume; when converted to energy content, the rates are 22 % for 2030 and 100 % for 2050. Co-firing rates between 2030 and 2050 have been calculated using interpolation, as illustrated in Figure C.10.

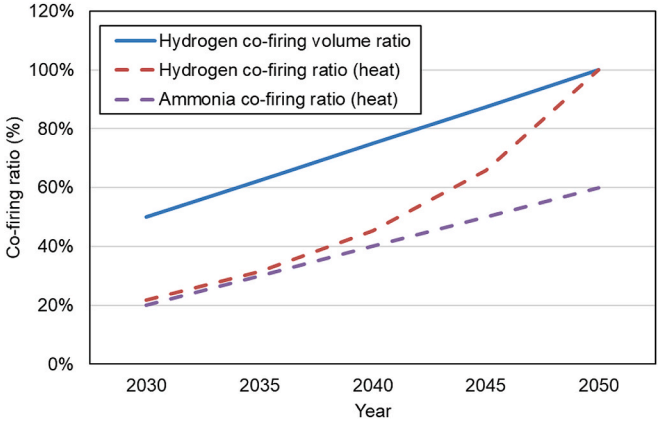


Fig. C.10. Co-firing ratio of green hydrogen with natural gas and green ammonia with coal

To fire blended hydrogen with natural gas, modifications to the existing components of natural gas turbines are necessary. We assume an 8.5 %

increase in the OCC of hydrogen gas turbines compared to natural gas-fired turbine power plants, based on alignment with leading turbine manufacturer designs and detailed, component-specific cost estimations [66]. For ammonia co-firing, we added infrastructure reinforcement costs to the OCC of coal-fired generators. An additional cost of $199/kW is assumed [52].

Carbon cost

In Korea, CO_2 emission allowances are traded under the Korea Emissions Trading Scheme (K-ETS). In 2023, the price was $9.7/$CO_2$ tonne; according to K-ETS Market Information Platform Korea Exchange Emission Trading System [105], it is expected to reach $50.4/$CO_2$ tonne in 2030 [106]. The International Energy Agency's World Energy Outlook 2024 presents three main scenarios reflecting different carbon cost pathways for the energy sector.

- *Stated Policies Scenario (STEPS).* This scenario assesses the current direction of the energy sector based on prevailing policy settings, market data, and technology costs. Market-based initiatives are expected to expand, with the CO_2 price for power generation under the EU ETS projected to reach just under $160/CO2 tonne by 2050 and $89/CO2 tonne in Korea.
- Announced Pledges Scenario (APS). This scenario explores the outcomes if governments fully and promptly achieve all declared energy and climate targets, including net zero ambitions. This scenario assumes full implementation of net zero emissions pledges, driving CO2 prices higher across all regions. By 2050 under this scenario, CO2 prices are expected to average $200/CO2 tonne in advanced economies, including Korea's.
- Net Zero Emissions by 2050 *(NZE) Scenario.* This scenario outlines a pathway to achieve net zero emissions by mid-century, aiming to limit global warming to 1.5 °C. Under this scenario, carbon pricing is expected to extend to all regions and most sectors, with prices reaching $250/$CO_2$ tonne in advanced economies, including Korea's.

In this study, the carbon price under the Announced Pledges Scenario is used across all scenarios.

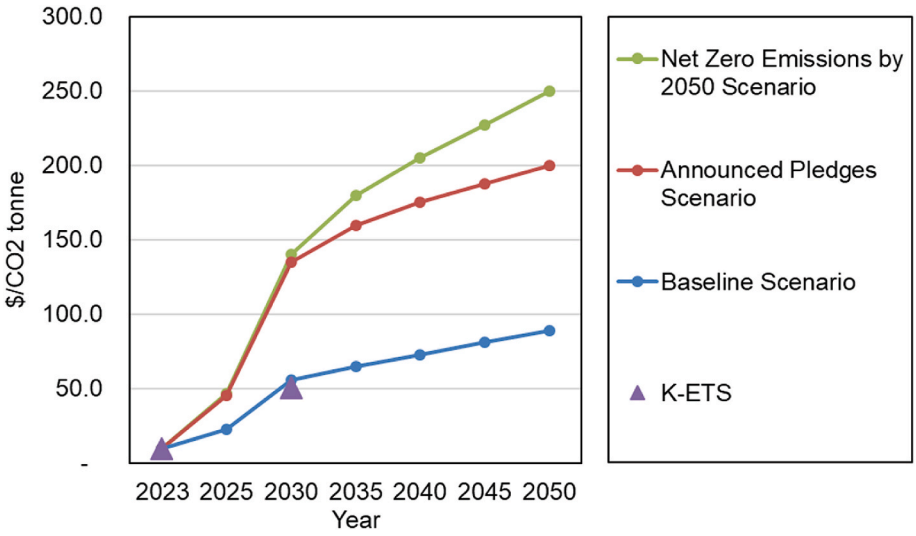


Fig. C.11. Carbon cost per emission under the Korea Emission Trading Scheme (K-ETS) and three IEA WEO scenarios [42]; Korea Exchange Emission Trading System [105]

Appendix D. LCOE for RE with energy storage and alternative LCOE metrics

LCOE for RE with energy storage

This analysis uses ATB costs for renewable energy paired with energy storage systems (ESS). When paired with ESS, the LCOE for wind technologies is projected to increase by 10–32 % depending on type, while the LCOE for solar PV increases significantly by 50–66 %. Two key factors drive this increase.

- The additional cost of battery storage, which adds approximately 50–60 % to overnight capital costs (OCC) compared to systems without ESS, as well as 20–25 % in fixed O&M (FOM) costs.
- The economic benefits of adding ESS do not increase proportionally with these added costs. One reason is that CF, which represents a long-term average over a plant's lifetime, does not fully capture the complexity of how an ESS operates. More specifically, CF overlooks short-term fluctuations in RE production and the frequent charging and discharging cycles of batteries in response to real-time electricity prices. As a result, the CF of ESS can appear lower than expected, since it does not reflect the full value provided by ESS during periods of grid imbalance or high price volatility.

Several additional examples shed further light on why the LCOE for RE resources increases when integrated with ESS, contrary to common expectations. First, under ATB assumptions, 75–100 % of the battery's energy is charged by the coupled PV system, which limits the portion of energy available to enhance the battery's CF. Second, not all energy stored in the ESS contributes significantly to improving its CF. When operating independently, the estimated ESS CF ranges from 8 % to over 40 %, depending on battery size and duration. However, this figure is diminished by constraints related to inverter and battery nameplate capacities and round-trip efficiency losses. Additionally, as referenced above, CF does not

account for seasonal variations in PV energy production or shifts in battery charging and discharging behavior influenced by fluctuating electricity prices. As a result, the ability of a DC-coupled solar PV system to capture energy, via ESS, that would otherwise be "clipped" (i.e., unused due to inverter constraints) produces only a minimal increase in CF. Consequently, relying solely on LCOE as a metric for energy planning may lead to the mistaken conclusion that investing in ESS is not advantageous.

It is also important to recognize that LCOE, as a cost metric, does not provide information about the various benefits generated by energy resources. Some studies have incorporated a wide range of benefits while examining the full costs of ESS. These analyses often depend on regional factors (e.g., regulatory environments) and the objectives of the organizations conducting the studies [107]. For instance, studies conducted from a regulatory perspective, particularly in U.S. states (e.g., California, Texas, Massachusetts, Colorado, Virginia, and North Carolina), frequently show that when considering broad benefits including social, economic, and utility system-wide advantages, the benefit-cost ratio (BCR) of ESS exceeds 1 [107]. At the organizational or individual consumer level, BCRs tend to be lower, ranging from 0.2 to 0.7 [108], even when accounting for benefits including avoided emissions, bill savings, and the value of lost load (VoLL); even so, these values help to significantly offset ESS installation costs [109].

Looking ahead, the economic landscape for ESS is expected to improve significantly. As costs decrease and benefits become more apparent, storage systems are projected to become increasingly cost-effective. This trend aligns with the growing necessity of ESS for integrating high penetrations of solar and wind energy, although widespread economic viability may not be realized until after 2030 Energy and Environmental Economics, Inc. [110]. In the U.S., solar-plus-storage systems have achieved cost-effectiveness in recent years with the support of the Investment Tax Credit (ITC) and may reach cost parity with new peaking power plants in the mid-2020s. Various support mechanisms are being planned to facilitate the introduction of renewable systems with ESS until they become inherently cost-competitive. While adding ESS inevitably increases LCOE for RE technologies, the long-term benefits are expected to outweigh the costs, particularly when considering milestones in 2030, 2035, and 2050, making ESS an increasingly promising technology over time.

Other metrics

While we have employed LCOE as the primary metric in this analysis due to its accessibility and widespread use, we acknowledge the value of alternative metrics that address its shortcomings. These metrics, discussed below, can offer a more holistic assessment of a technology's economic and system-level benefits, are quantifiable, and can be communicated effectively to diverse stakeholders.

- Value-adjusted LCOE (VALCOE), developed by the IEA, is an advanced metric that enhances the traditional LCOE by incorporating both the costs and the system value of electricity generation technologies. This comprehensive approach builds upon the average LCOE by integrating three critical elements: energy, capacity, and flexibility value. VALCOE offers a more nuanced assessment of competitiveness by considering technology-specific information and system characteristics, thereby providing a robust comparison across diverse operational profiles. The metric's strength lies in its ability to reflect the estimated value each technology contributes to the system, offering a more accurate representation of economic viability in complex energy landscapes. However, VALCOE is not without its limitations. Its scope closely aligns with System LCOE, potentially leading to redundancy in analysis. Additionally, while VALCOE represents a significant improvement over traditional LCOE, it may still fall short in fully capturing certain values, such as ancillary services, which play an increasingly important role in modern power systems [30].
- Levelized cost of storage (LCOS), a financial metric designed to assess the economic viability of energy storage systems, is analogous to the LCOE for power generation. LCOS calculates the average revenue per unit of electricity discharged necessary to recover the costs of building and operating a storage facility over its lifetime and duty cycle. Unlike LCOE, LCOS incorporates charging costs as a fuel equivalent and focuses on discharged rather than generated electricity, reflecting the unique characteristics of storage technologies that both consume and supply electricity to the grid [22]. LCOS's formulation allows for the inclusion of storage-specific components and terminology, providing a more accurate representation of costs compared to traditional LCOE calculations. This encompasses charging costs, augmentation, and replacement of storage and power equipment. The metric offers a comprehensive method for comparing various storage assets' true ownership and operational costs [31,111]. However, LCOS has limitations. It can oversimplify complex financial decisions by potentially omitting important factors and is sensitive to assumptions about project lifespan and financing rates. Furthermore, LCOS doesn't fully capture the additional value storage systems provide, such as grid services or avoided transmission costs, nor does it account for temporal variations in electricity prices. The metric's relative novelty also means that different organizations may use varying definitions, complicating cross-technology comparisons. Despite these challenges, LCOS remains a valuable tool for evaluating and comparing energy storage technologies, though it should be used in conjunction with other metrics for a more complete assessment [91].
- Levelized avoided cost of electricity (LACE) is defined as "a measure of what it would cost the grid to generate the electricity that would be displaced by a new generation project" [22].
- Levelized value of electricity (LVOE) is an advanced metric that quantifies the average economic value of each unit of electricity generated by a power plant throughout its operational lifespan. This concept emerged as a response to the recognized limitations of LCOE, particularly its failure to account for an asset's value to the broader energy system. LVOE is calculated by dividing the total present value of an asset's system contributions (including all services provided) by the present value of the electrical energy it produces. This approach offers a more holistic view of an asset's worth, contrasting with LCOE's focus on construction and operational expenses. Unlike LCOE, LVOE prioritizes the economic system value of the asset. It considers various services an asset provides to the grid, potentially including energy, capacity, and ancillary services. LVOE recognizes that an asset's value is not isolated but derives from its interaction with the broader energy system. Consequently, LVOE calculation necessitates sophisticated modeling to simulate system interactions and derive prices and service provisions. While LVOE addresses some of LCOE's shortcomings, it is important to note its limitations. LVOE values can vary significantly based on the specific energy system context, making comparisons across different systems challenging. The need for detailed system modeling can make LVOE calculations more complex and resource-intensive than LCOE. Similar to LCOE, LVOE doesn't account for the costs associated with providing the services it values, potentially skewing competitiveness assessments. Additionally, as a relatively new concept, methodologies for calculating LVOE may not be as standardized as those for LCOE [28].
- Net value of electricity (NVOE) is a value representing the total economic benefit of electricity generated from a power plant, accounting for both the revenue generated and the costs associated with its production. NVOE is typically calculated by subtracting the LCOE from the LVOE. This metric aims to assess the economic viability of energy technologies from a system perspective. Its strength lies in its comprehensive approach, considering both value and cost. However, its variations (e.g., "LCOE net of system value," which is the inverse of NVOE) and the potential for

differing results underscore the importance of using multiple metrics for a thorough evaluation of energy technologies. This approach ensures a more balanced assessment in the complex and evolving energy sector [28].

- System LCOE is a metric introduced by Ueckerdt et al. in Ref. [29,29]) to quantify the integration costs associated with variable renewable energy (VRE) technologies. It is expressed as Benchmark Price + LCOE - LVOE and is closely related to NVOE but with a sign difference and the inclusion of a benchmark price. The primary strength of system LCOE is its comprehensive assessment of a technology's economic viability within the broader energy system. It establishes a clear threshold that when system LCOE exceeds the benchmark price, the technology is not considered economically viable. This approach offers a more nuanced evaluation than traditional LCOE calculations by accounting for both costs and system value. However, the metric's reliance on a benchmark price introduces complexity. This price can be arbitrarily chosen or calculated using various methods, and it may change over time and between systems. This variability is both a strength, allowing for flexible analysis, and a limitation, requiring careful consideration when comparing results across studies or time periods [28].

While LCOE captures many critical variables that influence the competitiveness of constructing and operating a specific technology, it does not necessarily indicate which technology would provide the lowest-cost option for the electric grid at a specific location and time. Importantly, LCOE does not account for the economic value of a particular generation type to the system, making it less suitable for comparisons based on system-wide value. LCOE overlooks attributes that can vary significantly among energy resources, such as costs for ramping, startup, and shutdown, which are crucial for more detailed evaluations of a generator's cost and value to the grid system. Such in-depth analysis is typically conducted using electric sector models such as PLEXOS and the Regional Energy Deployment System (ReEDS) model. For example, Park et al. [1,70] present an economic analysis of decarbonizing Korea's power sector by 2035, utilizing detailed power system modeling that incorporates generation and transmission expansion planning with hourly dispatch.

Data availability

Data will be made available on request.

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